Instantons and Three-Manifolds

Lecture notes from the class of the same name taught by Irving Dai in Fall 2026, transcribed and expanded by Jemma Schroder

Acknowledgments: These notes were typeset with the assistance of Anthropic's Claude; any deviation from my handwritten notes are either expanding a shorthand or my own contribution in post-processing. Most mathematical content is directly from the lectures, everything else, as well as all mistakes, are my own. The style package is a variation on Andrew Lin's.

Contents

Part I: SU(2) Gauge Theory and Donaldson's Theorem

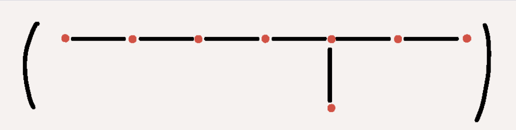
Lecture 1 // Aug 25 // Overview

Theorem 1.1 (Freedman)
Let Q be a unimodular symmetric bilinear form. Then there exists a unique simply connected topological 4-manifold X^4 such that $Q_X = Q$.

1. If Q is even, there exists a unique smooth X^4 up to homeomorphism.
2. If Q is odd, there exist at most two smooth X^4 up to homeomorphism, of which at most one is smoothable.

Theorem 1.2 (Donaldson)
If X^4 is a simply connected, smooth, closed 4-manifold and Q_X is definite, then Q_X is equivalent to $n\langle +1\rangle$ or $n\langle -1\rangle$.

Example 1.3
E_8 matrix:



$$\text{adj} = \begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & -2 \end{pmatrix}$$

Check: no change of basis takes this to $\pm I_8$.

$\Rightarrow$ E_8 is realized by a non-smoothable topological X^4.
$\Rightarrow$ the same holds for nE_8.

Over the next few weeks, we'll discuss the proof of Donaldson's theorem. After, we'll discuss Instanton Floer Homology (IFH) and some applications to three-manifolds, in particular, the homology cobordism group. The rest of this lecture will be to lay the groundwork for this.

3-Manifolds

Definition 1.4 (Homology Cobordism Group)

$$\Theta^3_{\mathbb{Z}} = \{\text{all } \mathbb{Z}\text{HS}^3\text{'s}\}/\text{homology cobordism}.$$

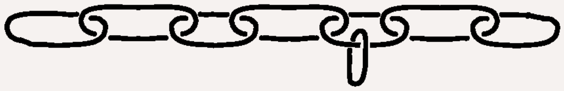
$Y_1 \sim Y_2$ are homology cobordant if there exists W^4, a homology $S^3 \times I$, such that $\partial W^4 \cong -Y_1 \sqcup Y_2$.

- The topological version is trivial.

Theorem 1.5 (Freedman)
If Y is a $\mathbb{Z}\text{HS}^3$, then $Y = \partial W$ for some $\mathbb{Z}\text{HB}^4$. In fact, $\pi_1 W = 1$.

Proof. Given a unimodular symmetric bilinear pairing Q, we can construct a smooth X^4 with boundary and intersection form Q by attaching 2-handles to B^4 according to Q:

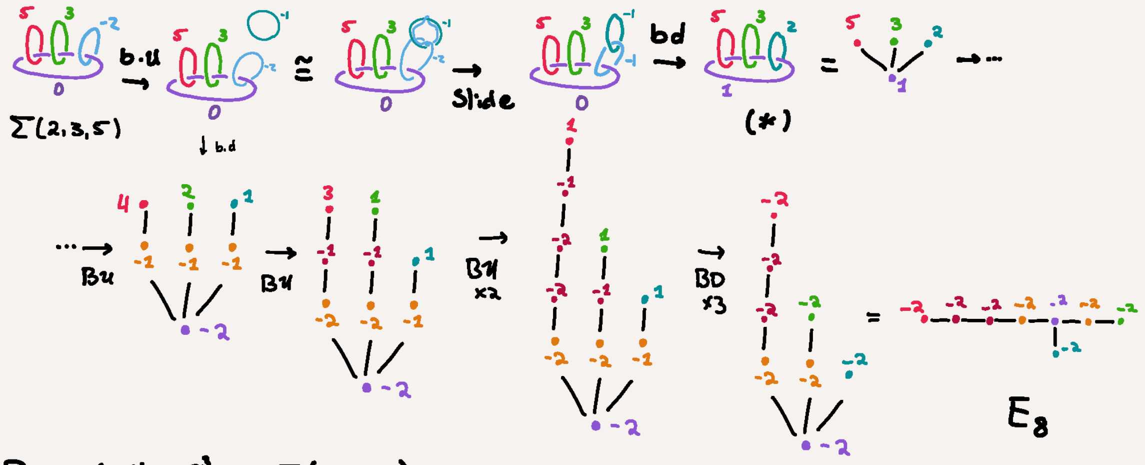
E_8:



For E_8, $\partial E_8 \cong$ Poincaré homology sphere $\cong \Sigma(2,3,5)$.
By Donaldson, $\Sigma(2,3,5)$ does not bound any smooth $\mathbb{Z}\text{HB}^4$

$$\Rightarrow [\Sigma(2,3,5)] \neq 1 \in \Theta^3_{\mathbb{Z}}.$$

Proof of boundary-diffeomorphism $\Sigma(2,3,5) \cong \partial E_8$: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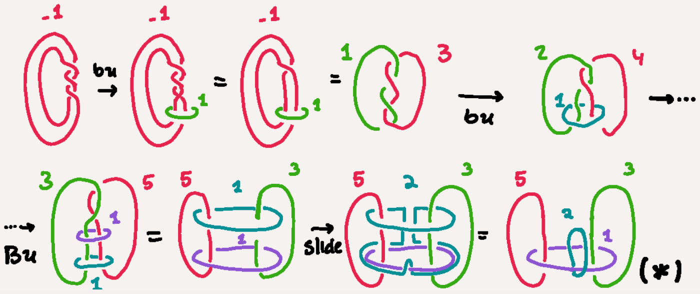
Poincaré Hx $S^3 \cong \Sigma(2,3,5)$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Remark 1.6

There are many $\mathbb{Z}\mathrm{HS}^3$'s.

Example 1.7

Brieskorn spheres: p, q, r pairwise coprime,

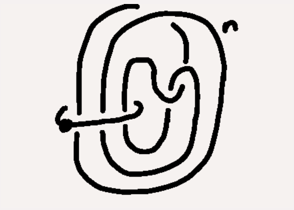
$$\Sigma(p,q,r) := S^5_\epsilon \cap \{x^p + y^q + z^r = 0\} \subseteq \mathbb{C}^6.$$

Example 1.8

$S^3_{1/n}(k)$ for all k.

Example 1.9

Mazur manifolds: 1-handle $\cup$ 2-handle such that the algebraic linking is ± 1, but the geometric linking is $\neq \pm 1$.



Question 1.10. *How large is* $\Theta^3_{\mathbb{Z}}$*?*

Theorem 1.11 (Furuta, FS)
$\{\Sigma(2,3,6n-1)\}$ is a $\mathbb{Z}^\infty$ subgroup of $\Theta^3_{\mathbb{Z}}$.

Theorem 1.12 (Daemi–Nozaki–Sato–Taniguchi)
Sometimes $\{S^3_{1/n}(k)\}$ are linearly independent.

This relies on a specific condition on IFH_*.

Note 1.13
$\Sigma(2,3,6n-1) = S^3_{1/n}(T_{2,3})$.

From concordance to homology cobordism:

$$\Sigma_2 : \mathcal{C} \longrightarrow \Theta^3_{\mathbb{Z}/2}, \qquad k \longmapsto \Sigma_2(k) = \text{double branched cover over } k,$$

is a homomorphism.

Question 1.14. *Is there a distinction between bounding a* $\mathbb{Z}\mathrm{HB}^4$ *and a* $\pi_1 = 0$ $\mathbb{Z}\mathrm{HB}^4$*? I.e., is there a* $\pi_1 = 1$ *homology cobordism from* $\Sigma(2,3,5)$ *to itself?*

Theorem 1.15 (Taubes)
No.

Theorem 1.16 (Daemi–Nozaki–Sato–Taniguchi)
Often, no. Cutting open the cobordism, we get a simply connected $\mathbb{Z}\mathrm{HB}^4$ bounding $Y\# -Y$.

Lecture 3 // Sept 1 // Examples & Hodge Theory

Last time:

A connection is a map $\nabla : \Gamma(E) \to \Gamma(T^*M \oplus E)$. Locally, connections take the form $\nabla = d + A$ where A is a connection 1-form matrix. Under change of coordinates, A transforms strangely, not by conjugation:

$$A \mapsto g^{-1}Ag + g^{-1}dg$$

where g is the bundle transition function. The curvature of a connection is $F_A := dA + A \wedge A$, and by Chern–Weil theory, the Chern classes are

$$\det\left(I - \frac{t}{2\pi}F_A\right).$$

Today, we'll explicitly compute one connection & its curvature to illustrate them in a toy example. This example will also generalize to help us understand the Yang–Mills moduli space.

Example: Connections on $\mathbb{CP}^1 = S^2$

Recall that $\mathbb{CP}^1 = \{[z : w] \mid z, w \in \mathbb{C}^2\}$ can be covered with 2 charts, the z chart with all points $[z : w]$ but $[0 : 1]$, and the w chart. The transition function is $w = \frac{1}{z}$.

Now, consider the $\mathbb{C}$-line bundle over $\mathbb{CP}^1$ defined by the bundle transition function $g(z) = z^n$. This is the degree-n $\mathbb{C}$-line bundle where $\mathbb{C}$-fibers get glued by z^n, viewing this as a 1×1 matrix over $\mathbb{C}$.

Fact 1.17

These are all the $\mathbb{C}$ line bundles over $\mathbb{CP}^1$.

What do connections on the degree-n line bundle look like? To define a connection, we must write down the connection's 1-form matrix A. Since the bundle is a line bundle, A is a 1×1 matrix over $\mathbb{C}$, i.e., a function $\mathbb{C} \to \mathbb{C}$.

Since A is purely local, we must define it in local coordinates. We'll consider the z chart first.

Proposition 1.18

$A := -n\dfrac{\bar{z}\,dz}{1 + |z|^2}$ defines a connection on the degree n $\mathbb{C}$-line bundle over $\mathbb{CP}^1$.

Proof. To check that A actually defines a connection, we must do the following:

1. look at A in the other chart
2. show that A extends smoothly over the point at ∞.

Each step requires its own coordinate change:

1. $A \rightsquigarrow g^{-1}Ag + g^{-1}dg$, the bundle change of coordinates

2. $w = \frac{1}{z}$, the manifold change of coordinates

1) Recall that the bundle transition function is $g(z) = z^n$.

Thus

$$\begin{aligned} g^{-1}Ag + g^{-1}dg &= z^{-n}\left(-n\frac{\bar{z}\,dz}{1+|z|^2}\right)z^n + z^{-n}d(z^n) \\ &= -n\frac{\bar{z}\,dz}{1+|z|^2} + \frac{nz^{n-1}}{z^n} \\ &= -n\frac{\bar{z}\,dz}{1+|z|^2} + \frac{n}{z} \end{aligned}$$

2) We now apply the manifold change of coordinates: $w = \frac{1}{z}$

$$dz = d\left(\frac{1}{w}\right) = \frac{-dw}{w^2}$$

So $-n\dfrac{\bar{z}\,dz}{1+|z|^2} + \dfrac{n}{z}$ becomes

$$\begin{aligned} \frac{-n\left(\frac{1}{w}\left(\frac{-1}{w^2}dw\right)\right)}{1+\left|\frac{1}{w}\right|^2} + nw\left(\frac{-1}{w^2}\right)dw &= -n\frac{dw}{w^3\left(1+\left|\frac{1}{w}\right|^2\right)} - n\frac{dw}{w} \\ &= \frac{-n\left(\frac{1}{w}\left(\frac{-1}{w^2}dw\right)\right)}{1+\left|\frac{1}{w}\right|^2} + nw\left(\frac{-1}{w^2}\right)dw \\ &= -n\frac{dw}{w^3\left(1+\left|\frac{1}{w}\right|^2\right)} - n\frac{dw}{w} \\ &= n\frac{dw}{w(1+|w|^2)} - n\frac{dw}{w} \\ &= n\left(\frac{1}{w(1+|w|^2)} - \frac{1}{w}\right)dw \\ &= n\left(\frac{1}{w(1+|w|^2)} - \frac{(1+|w|^2)}{w(1+|w|^2)}\right)dw \\ &= n\left(\frac{-|w|^2}{w(1+|w|^2)}\right)dw \\ &= \frac{-n\bar{w}\,dw}{1+|w|^2} \end{aligned}$$

this is a valid 1-form on the w-chart, and is defined at $z = \infty \leftrightarrow w = 0$.

Thus A defines a valid global 1-form. $\square$

Note that A itself does not extend as a smooth 1-form on $\mathbb{CP}^1$, since $n\frac{dw}{w(1+|w|^2)}$ is singular at $w = 0 \leftrightarrow z = \infty$. It is precisely the addition of the $g^{-1}dg = -n\frac{dw}{w}$ term from the bundle change of coordinates which resolves the issue. So A can be defined globally because of the "strange" bundle change-of-coordinates.

We'll now compute the curvature of this 1-form: $F_A = dA + A \wedge A$. Since $d^2 = 0$, $A \wedge A = 0$, we need only to compute dA.

$$\begin{aligned} dA = d\left(-n\frac{\bar{z}\,dz}{1+|z|^2}\right) &= -n\left(\frac{(1+|z|^2)d\bar{z} - \bar{z}(zd\bar{z}+\bar{z}dz)}{(1+|z|^2)^2}\right)\wedge dz \\ &= -n\frac{d\bar{z}\wedge dz}{(1+|z|^2)^2} \qquad d\bar{z} \\ &= dx - i\,dy, \quad dz = dx + i\,dy \\ &= -2ni\frac{dx\wedge dy}{(1+x^2+y^2)^2} \end{aligned}$$

Thus $F_A = -2ni\dfrac{dx\wedge dy}{(1+x^2+y^2)^2}$.

Using the formula for the Chern classes from Chern-Weil theory, we can compute the first Chern class; note that this is the expression of c_1 on the z-chart.

$$\Rightarrow c_1 = \frac{-1}{2\pi i}\left(-2ni\frac{dx\wedge dy}{(1+x^2+y^2)^2}\right) = \frac{n}{\pi}\frac{dx\wedge dy}{(1+x^2+y^2)^2}$$

Integrating over $\mathbb{CP}^1$, we get that

$$\begin{aligned} \int_{\mathbb{CP}^1} c_1 &= \frac{n}{\pi}\left(\int_{\mathbb{R}^2}\frac{dx\wedge dy}{(1+x^2+y^2)^2}\right) \\ &= \frac{n}{\pi}\int_0^{2\pi}\int_0^{\infty}\frac{r}{(1+r^2)^2}\,dr\,d\theta \\ &= \frac{n}{\pi}\int_0^{2\pi}\frac{1}{2}\,d\theta \\ &= n \end{aligned}$$

The previous computation leaves us with some questions.

1. If the final integral was taken over $\mathbb{R}^2$, where did $\mathbb{CP}^1$ come in to play? Why can't we choose another A defined only on $\mathbb{R}^2$?

2. In this example,

   $$\int_{\mathbb{CP}^1} c_1 = \int_{\mathbb{CP}^2} F_A = \int_{\mathbb{CP}^2} dA = \int_{\text{large disk}} dA = \int_{\text{large } S^1} A.$$

   Could we instead choose A to be compactly supported, so $\int = 0$? Or even choose $A = 0$?

The answer to both is a resounding no. The point here is that not every choice of A defines a connection on a $U(1)$ bundle over $\mathbb{CP}^1$. In this example, even though A is integrated over $\mathbb{R}^2$, we already checked that it extends, so the fact that A is defined globally comes first. A "knows" what bundle it lives on by "remembering" the singular behavior at $z = \infty$, since it must match the $\frac{n\,dw}{w}$ term to allow for cancellation.

If $A = 0$ or is compactly supported, $\dfrac{dw}{w(1+|w|^2)} = 0$, so $n\frac{dw}{w(1+|w|^2)} - n\frac{dw}{w} = -n\frac{dw}{w}$. For this term to be nonsingular, n must be 0, so the line bundle must be trivial. The upshot here is that the curvature integral determines the bundle via the behaviour near ∞.

A Review of Hodge Thery

Let (M, g) be an oriented Riemannian manifold. Recall that on each tangent space T_pM, we get an inner product pairing $\langle \cdot, \cdot \rangle_p$ which smoothly varies as p varies in M. This gives an inner product on each cotangent space T_p^*M which then induces an inner product on $\Lambda^k T^*M$ for each k as follows. Let $\{e_1, \dots, e_n\}$ be an oriented orthonormal basis of T_pM. Dualizing, we get a basis $\{e_1^*, \dots, e_n^*\}$ of T_p^*M which we may declare to be oriented and orthonormal; this declaration induces the inner product structure. For each fixed k, $\Lambda^k T^*M$ is $\binom{n}{k}$-dimensional. For each choice of k numbers between 1 and n, $\{i_1, \dots, i_k\} \subset \{1, \dots, n\}$, we get a basis vector of the form $e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$. Again, we declare this basis to be oriented and orthonormal, which induces the inner product structure on $\Lambda^k T^*M$. Of course, one must check that the resulting inner product doesn't depend on the original choice of oriented orthonormal basis for T_pM.

By abuse of notation, all of these inner products will be denoted by $\langle -, - \rangle$.

Now, let $n = \dim M$. On each $\Lambda^n T_p^*M$, there exists a unique oriented n-wedge with norm 1 via the above identification. This n-wedge is the volume form $e_1^* \wedge \cdots \wedge e_n^* =: d\mathrm{vol}_g$. Also that since $\Lambda^k T^*M$ has $\binom{n}{k}$ basis vectors, $\dim(\Lambda^k T^*M) = \dim(\Lambda^{n-k} T^*M)$; this is also true for differential forms. To exploit this symmetry, we may define the Hodge star.

Definition 1.19

The Hodge star is a map $\star : \Omega^k \to \Omega^{n-k}$ defined pointwise by (after fixing an orthonormal oriented basis $\{e_1, \dots, e_n\}$ of T_pM): for each $I \subset \{1, \dots, n\}$ with $|I| = k$, define $e_I^* := e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$.

For each basis vector e_I^*, the Hodge star sends e_I^* to $\star e_I^*$ via

$$\star e_I^* := \pm \underbrace{e_{j_1}^* \wedge \cdots \wedge e_{j_{n-k}}^*}_{e_J^*} \quad \text{where } J = \{1, \dots, n\} \setminus I.$$

The $\pm$ sign is chosen such that $e_I^* \wedge e_J^* = e_1^* \wedge \cdots \wedge e_n^*$, so $\star$ gives the complementary wedge of forms. To define $\star$ on non-basis vectors, we extend linearly.

Note that this definition is independent of the choice of orthonormal basis, but it does depend on the choice of metric. Furthermore, $\star$ is almost an involution… but not quite. To illustrate this behavior, consider $e_I = e_1 \wedge e_3$ and $e_J = -e_2 \wedge e_4$. Then

$$\begin{aligned} e_I \wedge e_J &= e_1 \wedge e_3 \wedge -e_2 \wedge e_4 \\ &= e_1 \wedge e_2 \wedge e_3 \wedge e_4 \\ &= d\mathrm{vol}_g \end{aligned}$$

so $\star(e_I) = e_J$. Similarly,

$$\begin{aligned} e_J \wedge e_I &= -e_2 \wedge e_4 \wedge e_1 \wedge e_3 \\ &= -e_2 \wedge -e_1 \wedge e_4 \wedge e_3 \\ &= e_1 \wedge -e_2 \wedge e_4 \wedge e_3 \\ &= e_1 \wedge -e_2 \wedge -e_3 \wedge e_4 \\ &= e_1 \wedge e_2 \wedge e_3 \wedge e_4 \\ &= d\mathrm{vol}_g \end{aligned}$$

Thus $\star(e_J) = e_I$, so in this case, $\star^2(e_I) = e_I$. However, this is a direct result of the number of swaps we had to perform to correctly order the basis vectors. In general, the total number of swaps depends on n and k: $(\star)^2 = (-1)^{k(n-k)}$.

Defining the Hodge star buys us two things. First, we can get an inner product structure on Ω^k from that on $\Lambda^k T^*M$, which was induced by the metric.

Definition 1.20

For $\alpha, \beta \in \Omega^k$, define

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge \star\beta \qquad \text{(since } \alpha \wedge \star\beta \text{is an n-form.)}$$

If α is an orthonormal basis element, $\alpha \wedge \star\alpha = d\mathrm{vol}_g$, so

$$\langle \alpha, \alpha \rangle = \int_M |\alpha|^2 \, d\mathrm{vol}_g.$$

Second, we get a formal adjoint of the exterior derivative. Recall that the exterior derivative is a map $d : \Omega^k \to \Omega^{k+1}$. A formal adjoint of d then must be a map $d^* : \Omega^k \to \Omega^{k-1}$.

Definition 1.21

The "codifferential" $d^* : \Omega^k \to \Omega^{k-1}$ is defined as the composition $d^* := (-1)^k \star d\star$:

$$d^* : \Omega^k \xrightarrow{\star} \Omega^{n-k} \xrightarrow{d} \Omega^{n-k+1} \xrightarrow{\star} \Omega^{k-1}.$$

Furthermore, $(d^*)^2 = 0$.

Lemma 1.22

d^* is the formal adjoint of d: $\langle d\alpha, \beta \rangle = \langle \alpha, d^*\beta \rangle$.

Proof. By Stokes' theorem and the product rule,

$$0 = \int_M d(\alpha \wedge \star\beta) = \int_M d\alpha \wedge \star\beta \pm \int_M \alpha \wedge d(\star\beta)$$

$$= \langle d\alpha, \beta\rangle \pm \int_M \alpha \wedge \star d^*\beta = \langle d\alpha, \beta\rangle \pm \langle \alpha, d^*\beta\rangle$$

$$\Rightarrow \langle d\alpha, \beta\rangle = \mp\langle \alpha, d^*\beta\rangle$$

□

We now have the tools to discuss the main point of Hodge theory (for our purposes).

Definition 1.23

The Hodge Laplacian is a map $\Delta : \Omega^k \to \Omega^k$ defined as

$$\Delta = (d + d^*)^2 = dd^* + d^*d.$$

k-forms $\alpha \in \Omega^k$ are called harmonic if they are in the kernel of the Hodge Laplacian, eg, $\Delta\alpha = 0$.

Lemma 1.24

α is harmonic iff $d\alpha = 0$ (closed) and $d^*\alpha = 0$ (co-closed).

Proof.

($\Leftarrow$) This obvious by the definition.

($\Rightarrow$) If $\Delta\alpha = 0$, $\langle \Delta\alpha, \alpha\rangle = 0$. Since $\Delta\alpha = dd^*\alpha + d^*d\alpha$,

$$0 = \langle \Delta\alpha, \alpha\rangle = \langle dd^*\alpha + d^*d\alpha, \alpha\rangle = \langle dd^*\alpha, \alpha\rangle + \langle d^*d\alpha, \alpha\rangle$$

$$= \langle d^*\alpha, d^*\alpha\rangle + \langle d\alpha, d\alpha\rangle = \langle (d + d^*)\alpha, (d + d^*)\alpha\rangle$$

$$\Rightarrow (d + d^*)\alpha = 0 \Rightarrow d\alpha = d^*\alpha = 0.$$

□

If α is harmonic, it is closed, so α defines a cohomology class $[\alpha] \in H^k_{dR}$. Furthermore, in local coordinates, $d^*d + dd^*$ is a 2nd order differential operator, since it acts like a 2nd derivative on the coefficients of basis vectors. We can interpret harmonicity in this context as cohomology classes which are norm minimizing. This gives the following lemma.

Lemma 1.25

α is harmonic iff α minimizes norm in its de Rham cohomology class.

Proof.

($\Rightarrow$) Let $\alpha \in H^k_{dR}(M; \mathbb{R})$. Every other element in this class will differ from α by an exact form, i.e., some $d\beta$ for some $\beta \in \Omega^{k-1}$. We will compute the norm of $\alpha + d\beta$.

$$\langle \alpha + d\beta, \alpha + d\beta\rangle = \langle \alpha, \alpha\rangle + \langle d\beta, d\beta\rangle + 2\langle \alpha, d\beta\rangle$$

We know that $\langle d\beta, d\beta\rangle \geq 0$, and $\langle \alpha, d\beta\rangle = \langle d^*\alpha, \beta\rangle = 0$ since α is harmonic.

Thus $\langle \alpha, \alpha\rangle \leq \langle \alpha + d\beta, \alpha + d\beta\rangle$.

(⇐) Assume α is norm minimizing. For all $\beta \in \Omega^{k-1}$, consider the ray $\langle \alpha + t\,d\beta, \alpha + t\,d\beta \rangle_{t\in[0,1]}$. I.e., we move α around in its cohomology class along some ray. Since α is norm minimizing, when $t = 0$, the derivative vanishes.

$$0 = \left.\frac{d}{dt}\right|_{t=0} \langle \alpha + t\,d\beta, \alpha + t\,d\beta \rangle = 2\langle \alpha, d\beta \rangle = 2\langle d^*\alpha, \beta \rangle$$

$$\Rightarrow d^*\alpha = 0$$

It remains to show $d\alpha = 0$ and that this minimizer is unique; the proof is left as an exercise. ☐

Theorem 1.26 (Hodge Theorem)

Define $\mathcal{H}^k$ to be the space of harmonic k-forms. There exists an isomorphism $\mathcal{H}^k \cong H^k_{dR}$ sending $\alpha \in H^k_{dR}$ to the norm minimizer in $[\alpha]$.

Lecture 4 // Sept 3 // ASD connections & Gauge

Last time:

To prove Donaldson's theorem, we need to study the space of anti-self-dual connections on M modulo gauge. Last time, we defined connections and and discussed the codifferential and the Hodge-Laplacian, and stated the main theorem of Hodge theory.

- $d^* = (-1)^k \star d\star$
- $\Delta := (d+d^*)^2 = dd^* + d^*d$
- $\Omega^k = \operatorname{im} d \oplus \operatorname{im} d^* \oplus \mathcal{H}^k$ where $\mathcal{H}^k := \ker d \cap \ker d^*$.

Today, we'll discuss self duality.

First, note that the statement of the Hodge decomposition given above isn't quite the same as from last time. It's not too difficult to verify that the decomposition $\Omega^k = \operatorname{im} d \oplus \operatorname{im} d^* \oplus \mathcal{H}^k$ are orthogonal, ie, have pairwise intersection $\{0\}$. However, it is harder to show that $\Omega^k = \operatorname{span}\{\mathcal{H}^k, \operatorname{im} d, \operatorname{im} d^*\}$. For this, you'll need some functional analysis, so we'll only verify the first part.

- If $\alpha \in \mathcal{H}^k \cap \operatorname{im} d$, $\Delta\alpha = 0$, so $\langle \alpha, \alpha\rangle = \langle \alpha, d\beta\rangle = \langle d^*\alpha, \beta\rangle = 0$.
- if $\alpha \in \operatorname{im} d$ and $\alpha \in \operatorname{im} d^*$, so $\langle \alpha, \alpha\rangle = \langle d\beta, d^*\gamma\rangle = \langle d^2\beta, \gamma\rangle = 0$.

Each cohomology class is in the kernel of d, and since $\ker(d) \cong \operatorname{im}(d) \oplus \mathcal{H}^k$, each class is in $\operatorname{im}(d) \oplus \mathcal{H}^k$. Therefore there can be only one harmonic representative. In particular, the different cohomology classes are affine cosets of $\operatorname{im}(d)$.

We'll now specialize to (M^4, g) and let $k = 2$. Since $n - k = 2$, the Hodge star gives an involution $\star : \Omega^2 \to \Omega^2$ which squares to the identity: $\star^2 = (-1)^{2(4-2)} = +1$. This means we can look at the positive and negative eigenspaces of $\star$ on Ω^2.

Definition 1.27

If $\star\alpha = \alpha$, α is called self-dual (SD). If $\star\alpha = -\alpha$, α is anti-self-dual (ASD). The subspace of self-dual 2-forms is denoted Ω^2_+, and the subspace of anti-self-dual 2-forms is Ω^2_-.

After choosing an orthonormal basis for $T_p^*M = \operatorname{span}\{e_1^*, e_2^*, e_3^*, e_4^*\}$, we get that

$$\Omega^2 = \operatorname{span}\{e_1 \wedge e_2, e_1 \wedge e_3, e_1 \wedge e_4, e_2 \wedge e_3, e_2 \wedge e_4, e_3 \wedge e_4\}.$$

To find the eigenspaces of $\star$, we'll compute $\star(e_i \wedge e_j$ for each basis element:

$$\star(e_1 \wedge e_2) = e_3 \wedge e_4, \qquad \star(e_1 \wedge e_3) = -e_2 \wedge e_4, \qquad \star(e_1 \wedge e_4) = e_2 \wedge e_3,$$

$$\star(e_2 \wedge e_3) = e_1 \wedge e_4, \qquad \star(e_2 \wedge e_4) = -e_1 \wedge e_3, \qquad \star(e_3 \wedge e_4) = e_1 \wedge e_2.$$

Therefore

$$\Omega^2_+ = \operatorname{span}\{e_1 \wedge e_2 + e_3 \wedge e_4,\ e_1 \wedge e_3 - e_2 \wedge e_4,\ e_1 \wedge e_4 + e_2 \wedge e_3\}$$

and

$$\Omega^2_- = \operatorname{span}\{e_1 \wedge e_2 - e_3 \wedge e_4,\ e_1 \wedge e_3 + e_2 \wedge e_4,\ e_1 \wedge e_4 - e_2 \wedge e_3\},$$

which splits the space of 2-forms as $\Omega^2 = \Omega^2_+ \oplus \Omega^2_-$

Further, note that if α is self-dual, $\int_M \alpha \wedge \alpha = \int_M \alpha \wedge \star\alpha = \langle \alpha, \alpha \rangle = \|\alpha\|^2$. Similarly, if α is anti-self-dual, $\int_M \alpha \wedge \alpha = \int_M \alpha \wedge - \star \alpha = -\int_M \alpha \wedge \star\alpha = -\langle \alpha, \alpha \rangle = -\|\alpha\|^2$, so self-dual 2-forms have positive self cup product, and anti-self-dual 2-forms have negative self cup product.

Classification of Bundles

Previously, we discussed vector bundless and connections on vector bundles. The bundles $E \xrightarrow{p} B$ had transition functions $g \in \mathrm{GL}_n(\mathbb{C})$, and connections $\nabla = d + A$ were defined locally as matricies with entries given by 1-forms. Now, we'll specify to SU(2)-connections.

Definition 1.28 (G-vector bundle)
Let G be a Lie group; think of it very explicitly as a Lie subgroup of $\mathrm{GL}_n(\mathbb{C})$. Then the Lie algebra $\mathfrak{g}$ is a subspace of $\mathrm{Mat}_n(\mathbb{C})$. We say that a vector bundle has structure group G if the transition functions $g \in \mathrm{GL}_n(\mathbb{C})$ are valued in the subgroup $G \subset \mathrm{GL}_n(\mathbb{C})$: $g \in G \subseteq \mathrm{GL}_n(\mathbb{C})$-valued. A vector bundle with this property is called a G-vector bundle.

Definition 1.29 (G-connection)
Given a G-vector bundle, a G-connection on E is a connection whose associated connection 1-form A is valued in $T^*M \otimes \mathfrak{g}$. This means that A is again a matrix of 1-forms, but the matrix must lie in Lie algebra. Even more concretely, given any tangent vector, we may feed it to A. This results in a matrix of numbers–a literal element of $\mathrm{Mat}_n(\mathbb{C})$–which is required to be in $\mathfrak{g} \subseteq \mathrm{Mat}_n(\mathbb{C})$).

Exercise 1.30
Check that the definition of a G-connection is preserved by coordinate change

$$A \mapsto g^{-1}Ag + g^{-1}dg \quad \text{(as long as } g \in G\text{)}$$

e.g., if $A \in \mathfrak{g}$, $g^{-1}Ag + g^{-1}dg \in \mathfrak{g}$:

$\hookrightarrow$ $\mathfrak{g}$ closed under conjugation with $g \in G$

$\hookrightarrow$ dg in $T_eG \rightsquigarrow$ must move to e before evaluating, so precompose with g^{-1}

Exercise 1.31
If $\nabla = d + A$ is a G-connection, $F_A = dA + A \wedge A \in \Lambda^2 T^*M \otimes \mathfrak{g}$ at each point, i.e., for any v, w, feed in v, w to 2-form entries in matrix, get a matrix in $\mathfrak{g} \in \mathrm{Mat}_n(\mathbb{C})$.

F_A can equivalently be written as

$$F_A \in \Gamma\big(\Lambda^2 T^*M \otimes \underbrace{\operatorname{ad}(E)}_{\text{adjoint bundle}}\big)$$

Before, $F_A \in \Gamma(\Lambda^2 T^*M \otimes \operatorname{End}(E))$. $\operatorname{ad}(E) \subset \operatorname{End}(E)$ is the subbundle of $\operatorname{End}(E)$ with all transition functions in $\mathfrak{g}$; $\mathfrak{g}$ is closed under conjugation, so transition functions on $\operatorname{End}(E)$ restrict to those on $\operatorname{ad}(E)$.

We write $\Omega^2(\operatorname{ad} E) := \Gamma(\Lambda^2 T^*M \otimes \operatorname{ad}(E))$.

In our setting, there will be 2 G's which are relevant:

① $G = \mathrm{U}(1)$ = complex line bundles $\subset \mathrm{GL}_1(\mathbb{C})$

so g is S^1 valued

so tangent space at 1 in $\mathrm{U}(1)$ is imaginary axis $\Rightarrow A \in i\,\Omega^1(M;\mathbb{R})$

② $G = \mathrm{SU}(2) \subseteq \mathrm{GL}_2(\mathbb{C})$

so g looks like $\begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}$, $|a|^2 + |b|^2 = 1$

so A is locally in $\begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & -\alpha \end{pmatrix}$, α is purely imaginary 1-form

$\mathfrak{su}(2)$ = trace-free skew-hermitian matrices.

How to classify these G-bundles?

Recall Chern–Weil: For $\mathrm{U}(1)$ bundles, only 1 chern class c_1.

Theorem 1.32

$c_1 \in H^2(M;\mathbb{Z})$ classifies $\mathrm{U}(1)$ bundles $/M$.

For $\mathrm{SU}(2)$, have c_1 and c_2. If F_A is 2×2, check that:

$$c_1 = \frac{-1}{2\pi i}\operatorname{tr}(F_A) \quad \text{(coeff. of } t) \rightsquigarrow \text{ in } \det\big(I - \tfrac{t}{2\pi i}F_A\big)$$

$$c_2 = \frac{-1}{4\pi^2}\det(F_A) \quad \text{(coeff. of } t^2)$$

Specifically, for an $\mathrm{SU}(2)$ connection, we know that

$$c_1 = \frac{-1}{2\pi i}(\alpha + -\alpha) = 0$$

$$c_2 = \frac{-1}{4\pi^2}\left(\alpha \wedge -\alpha - \beta \wedge -\bar{\beta}\right)$$

For an $\mathrm{SU}(2)$ bundle, also have that

$$c_2 = \frac{1}{8\pi^2}\operatorname{Tr}(F_A \wedge F_A)$$

$$\left(F_A \wedge F_A = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & -\alpha \end{pmatrix} \wedge \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & -\alpha \end{pmatrix} = \begin{pmatrix} \alpha \wedge \alpha - \bar{\beta} \wedge \beta & * \\ * & -\beta \wedge \bar{\beta} + \alpha \wedge \alpha \end{pmatrix}\right)$$

Theorem 1.33

SU(2) bundles over M^4 are classified by c_2.

Not true in general: $\pi_k(B\,\mathrm{SU}(2)) \neq 1\ \forall\ k > 4$.

In particular, a (closed oriented) 4-manifold has $\mathbb{Z}$-many SU(2) bundles, since $c_2 \in H^4(M;\mathbb{Z}) \cong \mathbb{Z}$.

Definition 1.34

Given an SU(2) bundle, the "instanton number" = "2^{nd} chern number" of E is

$$k = -\int_M c_2 \quad (= -\langle [M], c_2\rangle, \text{ convention})$$

Splittings of Bundles

Let E be an SU(2) bundle. When does E split as $E = L_1 \oplus L_2$?, $L_i = \mathrm{U}(1)$ bundle, transition functions h split as $h = \begin{pmatrix} h_{11} & 0 \\ 0 & h_{22} \end{pmatrix}$.

If E splits, it's called a split bundle, $L_1 \oplus L_2$ is a "splitting." If E is split, a split connection is locally of the form $A = \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{pmatrix}$.

If E is an SU(2) bundle, any splitting into U(1) bundles must be of the form $L \oplus \bar{L}$, since $\alpha_1 = -\alpha_2$

$$\Rightarrow \text{a split SU(2) connection looks like } A = \begin{pmatrix} \alpha & 0 \\ 0 & -\alpha \end{pmatrix}$$

Question 1.35. *If E splits, how many splittings?*

If E splits, $F_A = \begin{pmatrix} d\alpha & 0 \\ 0 & -d\alpha \end{pmatrix}$ since $A \wedge A = 0$.

$$\Rightarrow \text{if } E \text{ splits as } L \oplus \bar{L}, \quad c_2(E) = c_1(L) \cup c_1(\bar{L}) = c_1(L) \cup -c_1(L)$$

$$\Rightarrow \text{if } E \text{ splits, determined by 1 line bundle, which is determined by one } H^2 \text{ class}$$

$$\Rightarrow k = -\langle c_2, [M]\rangle = -\langle c_1 \cup -c_1, [M]\rangle = \langle c_1 \cup c_1, [M]\rangle$$

Lemma 1.36

Let E be an SU(2) bundle with instanton # k over M^4. Then E splits iff there exists $\alpha \in H^2(M, \mathbb{Z})$ such that $\int_M \alpha^2 = k$, and the # splittings = $\frac{1}{2}$# such α (if H^2 torsion free).

The Punchline:

Definition 1.37

Let (M^4, g) be a Riemannian 4-manifold and E be an SU(2) bundle over M (will usually consider $k = 1$).

$$\mathcal{A} := \{\text{SU(2) connections on } E\}$$

this is an ∞-dimensional affine space over $\Gamma(T^*M \otimes \operatorname{ad} E)$.

$$\mathcal{G} := \{\text{SU(2) gauge group}\} = \Gamma(\operatorname{Aut}(E))$$

section in $\operatorname{End}(E)$ valued in $\operatorname{ad}(E)$ (locally).

The gauge group acts on the space of connections: $\mathcal{G} \curvearrowright \mathcal{A}$ by

$$g \curvearrowright A = g^{-1}Ag + g^{-1}dg$$

$\mathcal{G}$ is similar to $[M, \text{SU}(2)]$. It's not quite; this is only true if E is trivial.

Definition 1.38

A connection $\nabla = d + A$ is self-dual if $F_A = \star F_A$, anti-self-dual if $F_A = - \star F_A$

(extend $\star$ to act on $\Gamma(\Lambda^2 T^*M \otimes \operatorname{ad}(E))$ by acting on just the 2-forms part; e.g., $\star$ of every entry of matrix.)

(Really, should say "connection with self-dual curvature.") Check: property of being self dual is preserved by gauge transformations: $F_A \mapsto g^{-1}F_A g$, $\star$ commutes with conjugation. E.g., the property of being SD is invariant under $\mathcal{G}$.

Definition 1.39

The (A)SD Moduli Space is

$$\mathcal{M} := \{\text{(anti)-self-dual SU(2) connections}\}/\mathcal{G} \subset \mathcal{A}/\mathcal{G}$$

Lecture 5 // Sept 8 // Yang Mills Functional & Equations

Last time:
We defined the following objects: (M^4, g), E an SU(2) bundle over M.

$$\mathcal{A} := \text{space of } \mathrm{SU}(2) \text{ connections } \{\nabla = d + A\}, \text{ this is an affine space over } \Gamma(T^*M \otimes \operatorname{ad} E).$$

$$\mathcal{G} = \mathrm{SU}(2) \text{ gauge group} = \Gamma(\operatorname{Aut} E), \qquad g \curvearrowright A, \qquad A \mapsto g^{-1}Ag + g^{-1}dg.$$

$\mathcal{M}$ = Moduli space of Yang Mills connections with self dual curvature, i.e.,

$$\mathcal{M} = \{\mathrm{SU}(2) \text{ connections } \nabla = d + A \text{ such that } F_A = \star F_A \text{ (self dual)}\} / \mathcal{G}.$$

This condition of having self-dual curvature is actually a specific instance of a larger phenomena, which we'll discuss today to put A(SD) connections in context.

The Yang Mills Functional

Recall that the Lie algebra of SU(2), $\mathfrak{su}(2)$, is defined as the subset of 2×2 complex matrices which are trace-free and skew-hermitian.

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} ix & -y+iz \\ y+iz & -ix \end{pmatrix} \,\middle|\, (x,y,z) \in \mathbb{R}^3 \right\}$$

In particular, this definition clearly shows that SU(2) is in bijection with $\mathbb{R}^3$. $\mathfrak{su}(2)$ can be endowed with an inner product given by $\langle A, B\rangle = \operatorname{tr}(AB^\dagger) = -\operatorname{tr}(AB)$, which is the same inner product as the one on $\mathbb{R}^3$ (the dot product).

$$\left\langle \begin{pmatrix} ix & -y+iz \\ y+iz & -ix \end{pmatrix}, \begin{pmatrix} ip & -q+ir \\ q+ir & -ip \end{pmatrix} \right\rangle = 2(xp + yq + zr) = 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

This inner product on $\mathfrak{su}(2)$ can be extended to an L^2 inner product on $\Omega^i(\operatorname{ad} E) = \Gamma(\Lambda^i T^*M \otimes \operatorname{ad} E)$. Given two sections F and G, their inner product is defined as

$$\langle F, G\rangle_{L^2} = -\int_M \operatorname{tr}(F \wedge \star G)$$

where the inner product and trace is taken pointwise for each $p \in M$, giving a 4-form at each point, i.e., a function of p which we can integrate over M. By taking $G = F$, the norm of F is thus $\|F\|^2 = \langle F, F\rangle_{L^2}$.

Using this inner product, we may now define the Yang-Mills Functional.

Definition 1.40 (Yang-Mills Functional)

$$\mathrm{YM} : \{\mathrm{SU}(2)\text{-connections}\} \to \mathbb{R}$$
$$\nabla \longmapsto \mathrm{YM}(\nabla) := \langle F_A, F_A \rangle_{L^2} = -\int_M \mathrm{tr}(F_A \wedge \star F_A) = -\int_M \|F_A\|^2 \, d\mathrm{vol}_M$$

Heuristically, $\mathrm{YM}(\nabla)$ measures the "strength" of curvature of ∇. For some connection, what does it mean to be a critical point of the Yang-Mills functional? Recall that the space of connections is affine over $\Gamma(T^*M \otimes \mathrm{ad}\, E)$. A fixed connection ∇ is a critical point of YM if every nearby connection ∇' has larger or smaller curvature. Any such ∇' can be written as $\nabla' = \nabla + tB$, i.e., a small perturbation of ∇ by some section $B \in \Gamma(T^*M \otimes \mathrm{ad}\, E)$.

In terms of F_A, the curvature of ∇' becomes

$$F_{A+tB} = d(A+tB) + (A+tB) \wedge (A+tB) = F_A + t(dB + A \wedge B + B \wedge A) + t^2 B \wedge B.$$

Define $D_A B := dB + A \wedge B + B \wedge A$.

The derivative $\left.\frac{d}{dt}\right|_{t=0} \mathrm{YM}(\nabla + tB)$ then becomes

$$\left.\frac{d}{dt}\right|_{t=0} \mathrm{YM}(\nabla + tB) = \left.\frac{d}{dt}\right|_{t=0} \langle F_{A+tB}, F_{A+tB} \rangle$$
$$= \left.\frac{d}{dt}\right|_{t=0} \langle F_A + tD_A B + t^2 B \wedge B,\ F_A + tD_A B + t^2 B \wedge B \rangle$$
$$= 2\langle D_A B, F_A \rangle.$$

Therefore ∇ is a critical point of YM iff this pairing is 0 for all B.

Before we continue, we should discuss the appearance of $D_A B$.

Interlude: More about connections & bundles

Recall the usual deRham complex,

$$\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots$$

Now, take a vector bundle $E \to B$ and a fixed connection ∇ on E. We have already described ∇ as a map $\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E)$. Can we get an analogous complex to the deRham complex?

To do so, we must generalize the spaces & the maps. Recall that the Ω^i in the deRham complex are defined to be sections of the bundle $\Lambda^i T^*E \to M$.

Definition 1.41

E-valued i-forms are sections of the bundle $\Lambda^i T^*M \otimes E$:

$$\Omega^i(E) = \Gamma(\Lambda^i T^*M \otimes E).$$

Note that when $i = 0$, we recover $\Omega^0(E) = \Gamma(\Lambda^0 T^*M \otimes E) = \Gamma(E)$, and when $i = 1$, we get

$\Omega^1(E) = \Gamma(\Lambda^1 T^*M \otimes E)$. So as defined above, a connection gives a map $\nabla : \Omega^0(E) \to \Omega^1(E)$. We extend the definition of ∇ using the Leibniz rule, as follows.

Definition 1.42

$$\nabla : \Omega^i(E) \to \Omega^{i+1}(E)$$

$$\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^i \alpha \wedge \nabla s$$

where α is an i-form and s a section of E, so $(\alpha \otimes s) \in \Omega^i(E)$.

Note that unlike in the deRham complex, $\nabla^2 \neq 0$. However, measuring the failure of this fact is very useful.

Lemma 1.43
$\nabla^2(s) = F_A s$.

Proof. Choose basis sections $\{e_1, \ldots, e_n\}$ by choosing a trivialization of the bundle. Then $s = \sum s_i e_i$, where s_i are functions.

Recall $\nabla = d + A$. Therefore

$$\nabla^2 s = \nabla(\nabla s) = \nabla\left(\sum_{i,j} ds_i \otimes e_i + s_i A_{ji} e_j\right) \quad \text{(Einstein but we have a single sum here)}$$

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + \underbrace{\nabla(s_i A_{ji}) e_j \wedge ds_i e_i}_{\nabla \text{ of 2nd term}}$$

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + \big(ds_i \wedge A_{ji} + s_i dA_{ji}\big) e_j - (s_i A_{ji}) \wedge (A_{kj} e_k)$$

$$= s_i\, dA_{ji} e_j + s_i A_{j\ell} A_{\ell i} e_j$$

$$= dA + A \wedge A.$$

□

The upshot here is that curvature is the obstruction for the differential on the $\Omega^i(E)$ complex to square to 0.

We'll now consider the bundle $\operatorname{End}(E)$. Given a connection $\nabla = d + A$ on E, we can define a connection on $\operatorname{End}(E)$ using ∇. We do so locally.

Definition 1.44
A connection D_A on $\operatorname{End}(E)$ is locally of the form

$$D_A B := dB + AB - BA$$

where A is a matrix of 1-forms and B is a section of $\operatorname{End}(E)$, i.e., a matrix valued function.

This definition can be motivated in two ways. First, connections on E are identified with matrices, which act on vectors by multiplication and other matrices via the Lie bracket. The dB term is the vector part, i.e., the usual entry-wise derivative, and the $AB - BA$ term records the linear action of A on B.

The second way to motivate this is by decomposing $\operatorname{End}(E)$ as $\operatorname{End}(E) = E \otimes E^*$, and checking that tensoring ∇ with the connection on E^* obtained by dualizing gives the same thing.

Now that we have a connection on $\operatorname{End}(E)$, we again get a map $D_A : \Omega^0(\operatorname{End} E) \to \Omega^1(\operatorname{End} E)$, which we can also extend to a map $D_A : \Omega^i(\operatorname{End} E) \to \Omega^{i+1}(\operatorname{End} E)$.

Exercise 1.45
Use the Leibniz rule to extend D_A to a map $D_A : \Omega^i(\operatorname{End} E) \to \Omega^{i+1}(\operatorname{End} E)$.

Further, if A is a $\mathrm{SU}(2)$ connection, B is $\mathfrak{su}(2)$-valued, so $[A, B]$ is also $\mathfrak{su}(2)$-valued. Therefore we can restrict $\operatorname{End}(E)$ to $\operatorname{ad}(E)$ and set the complex

$$\Omega^0(\operatorname{ad} E) \xrightarrow{D_A} \Omega^1(\operatorname{ad} E) \to \cdots \qquad (*)$$

Lemma 1.46
For $B \in \Omega^1(\operatorname{ad} E)$, $D_A B$ as defined above is the $D_A B$ defined. $\boxed{\text{!! Sign issue?}}$

Proof. First, any $B \in \Omega^1(\operatorname{ad} E)$ can be expressed as a linear combination of 1-forms ω with coefficients constant $\mathfrak{su}(2)$ matrices C. Thus we can assume $B = \omega C$.

Then $D_A B = D_A(\omega C)$

$$\underset{\substack{\text{Leibniz}\\ \text{and } \omega \text{ is a 1-form}}}{=} d\omega\, C - \omega \wedge \nabla C$$

$$\underset{(\text{since } C \text{ const})}{=} d\omega\, C - \omega \wedge (dC + AC - CA)$$

$$\underset{\substack{(\text{since } \omega, A \text{ both}\\ \text{1-forms,}\\ \omega\wedge A = -A\wedge\omega)}}{=} d\omega\, C - \omega \wedge [A, C] = d\omega\, C - \underbrace{(\omega \wedge A)}\, C + \omega C \wedge A$$

$$= d\omega\, C - (-A \wedge \omega) C + \omega C \wedge A$$

$$= d(\omega C) + A \wedge (\omega C) + (\omega C) \wedge A$$

$$= dB + A \wedge B + B \wedge A.$$

□

Note that this proof, applied to $\omega = i$-form instead, shows in general that the map

$$D_A B : \Omega^i(\operatorname{End} E) \to \Omega^{i+1}(\operatorname{End} E) \text{ is}$$

$$D_A B = dB + A \wedge B - B \wedge A, \text{ for } i \text{ even,}$$

$$\text{and } D_A B = dB + A \wedge B + B \wedge A, \text{ for } i \text{ odd.}$$

Now, seemingly from nowhere, we'll define the Bianchi Identity – this will be incredibly useful later.

Theorem 1.47 (Bianchi Identity)
If A is a fixed connection, $D_A F_A = 0$, $F_A \in \Omega^2(\mathrm{ad}(E))$.

Proof. Since $F_A = dA + A \wedge A$,

$$D_A F_A = dF_A + A \wedge F_A - F_A \wedge A$$

$$= d(dA + A \wedge A) + A \wedge (dA + A \wedge A) - (dA + A \wedge A) \wedge A$$

$$= \big(d^2 A + d(A \wedge A)\big) + \big(A \wedge dA + A \wedge A \wedge A\big) - \big(dA \wedge A + A \wedge A \wedge A\big)$$

$$= d^2 A + d(A \wedge A) + A \wedge dA + A \wedge A \wedge A - dA \wedge A - A \wedge A \wedge A$$

$$\underset{\substack{\text{(since } d^2A=0 \text{ and} \\ dA\wedge A=-A\wedge dA)}}{=} (dA \wedge A + A \wedge dA) + A \wedge dA - dA \wedge A \qquad \boxed{\text{Sign issue?}}$$

$$= d^2 A + dA \wedge A + A \wedge dA + A \wedge dA + A \wedge A \wedge A - dA \wedge A - \underline{A \wedge A \wedge A}$$

$$= 0.$$

□

Now, let $D_A^* := \star D_A \star$. This gives us a codifferential

$$D_A^* : \Omega^2(\mathrm{ad}\, E) \to \Omega^1(\mathrm{ad}\, E).$$

We are now ready to return to discussing the critical points of the Yang-Mills Functional.

Yang-Mills Connections

We have seen that A is a critical point of YM iff

$$\langle D_A B, F_A \rangle_{L^2} = 0 \quad \text{for all } B \in \Omega^1(\mathrm{ad}\, E)$$

$$\Leftrightarrow \langle B, D_A^* F_A \rangle_{L^2} = 0$$

$$\Leftrightarrow D_A^* F_A = 0.$$

Definition 1.48
A connection such that $D_A^* F_A = 0$ is called a Yang-Mills connection. These connections are the critical points of the YM functional.

Note the similarity with Hodge theory! Last week, we saw that the critical points of L^2 norm on forms are precisely the harmonic forms: α for which $d\alpha = d^*\alpha = 0$.

If a connection A is Yang-Mills, then $D_A^* F_A = 0$. And by the Bianchi identity, we always have that $D_A F_A = 0$. Therefore YM connections can be thought of as harmonic forms; the slogan here is that "YM theory is the SU(2) version of Hodge theory."

The one caveat is that in general, $D_A^* F_A$ is difficult to understand. In Hodge theory, d always exists, but here, both D_A^* and F_A depend on A.

It turns out that some YM connections satisfy a simpler condition, which also guarantees that they are global minima/maxima.

Lemma 1.49
1. If A is SD or ASD, then $D_A^* F_A = 0$.
2. In fact, (A)SD are the global minima/maxima of the YM functional.

Proof.

1. This is immediate from Bianchi: since $D_A F_A = 0$ and $D_A^* = \star D_A \star$, if $F_A = \pm \star F_A$, then
$$D_A^* F_A = \star D_A \star F_A = \star D_A \pm F_A = \pm \star D_A F_A = 0.$$
2. For $F_A \in \Omega^2(\operatorname{ad} E) = \Omega^2_+(\operatorname{ad} E) \oplus \Omega^2_-(\operatorname{ad} E)$, it splits as $F_A = F_A^+ + F_A^-$, where $\langle F_A^+, F_A^- \rangle_{L^2} = 0$.

   Thus
$$\begin{aligned} \mathrm{YM}(\nabla) = \langle F_A, F_A \rangle_{L^2} &= -\int_M \operatorname{tr}(F_A \wedge \star F_A) \\ &= \langle F_A^+ + F_A^-,\ F_A^+ + F_A^- \rangle_{L^2} \\ &= \langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^-, F_A^- \rangle_{L^2} + \langle F_A^+, F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2} \\ &= \|F_A^+\|_{L^2}^2 + \|F_A^-\|_{L^2}^2. \end{aligned}$$

   Now, recall the instanton number k defined as
$$k = -\langle c_2, [M] \rangle = \frac{-1}{8\pi^2} \int_M \operatorname{tr}(F_A \wedge F_A).$$

   Since $F_A = \star(\star F_A)$,
$$\begin{aligned} k &= \frac{1}{8\pi^2} \langle F_A, \star F_A \rangle_{L^2} \\ &= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-,\ \star(F_A^+ + F_A^-) \rangle_{L^2} \\ &= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-,\ F_A^+ - F_A^- \rangle_{L^2} \\ &= \frac{1}{8\pi^2} \Big(\langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^+, -F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2} + \langle F_A^-, -F_A^- \rangle_{L^2} \Big) \\ &= \frac{1}{8\pi^2} \Big(\|F_A^+\|_{L^2}^2 - \|F_A^-\|_{L^2}^2 \Big). \end{aligned}$$

   Therefore $8\pi^2 k = \|F_A^+\|_{L^2}^2 - \|F_A^-\|_{L^2}^2$. Combining this with $\mathrm{YM}(\nabla) = \|F_A^+\|_{L^2}^2 + \|F_A^-\|_{L^2}^2$, we see that

   if $k > 0$, then $\mathrm{YM} \geq 8\pi^2 k$, with equality when $F_A^- = 0$, i.e., when A is self-dual. On the other hand, if $k < 0$, then $\mathrm{YM} \geq 8\pi^2(-k)$ and equality is realized when $F_A^+ = 0$, so A is anti-self-dual. $\square$

Lecture 6 // Sept 10 // $U(1)$ Yang-Mills Theory

Last time:

We defined

$$D_A^*:\Omega^1(\operatorname{End} E)\to\Omega^1(\operatorname{End} E),\qquad D_A^*=\star D_A\star .$$

The Yang–Mills equation: $D_A^*F_A=0$.

Connections which satisfy the YM equation are called YM connections, and are the critical points of the Yang–Mills functional

$$YM(\nabla)=\int_M\|F_A\|_{L^2}^2\,d\text{vol}.$$

When the instanton number k is:

- $k>0$: self dual connections are the absolute minima
- $k<0$: anti self dual connections are the absolute maxima

Today: we'll make the connection with Hodge theory explicit.

Some remarks:

1. Can check that the Yang–Mills equation is invariant under gauge transformations.
2. There is an easier way to motivate D_A:

$$\text{Recall}\quad D_AB=dB+\underbrace{AB-BA}_{\text{Lie bracketing against }A}$$

More concretely: if $B\in\Gamma(\operatorname{End} E)$, $s\in\Gamma(E)$, should expect some sort of Leibniz rule:

$$\nabla(Bs)=(?)s+B(\nabla s)$$

and compute what the covariant derivative of B ($=(?)$) by completing the rest.

$U(1)$ Yang-Mills Theory

Fix a U(1) bundle L over (M^4,g) a Riemannian four manifold. To find the Yang-Mills connections on L, we must find $\mathcal{A}$, the action of $\mathcal{G}$, the explicit spaces and maps in the cochain complex. With those in hand, we can recast the Bianchi Identity and the Yang Mills condition in the $U(1)$ setting.

The space of U(1) connections is $\mathcal{A}=\{\nabla=d+ia\}$ where a is locally an element of $\Omega^1(M)$. Since $\dim \mathrm{U}(1)=1$, our matrices are 1×1, i.e., functions. For L a line bundle, End(L) is a trivial bundle, so the gauge group is smooth maps of $M\to S^1$, $\mathcal{G}=\operatorname{Aut}L=\operatorname{Maps}(M,S^1)$. Then $\mathcal{G}$ acts on $\mathcal{A}$ by $a\mapsto g^{-1}ag+g^{-1}dg$, but since U(1) is abelian, $g^{-1}ag=a$ and this action simplifies to $a\mapsto a+g^{-1}dg$.

We now consider the cochain complex

$$D_A:\Omega^i(\operatorname{ad}L)\to\Omega^{i+1}(\operatorname{ad}L).$$

Since $\operatorname{Aut} L = \operatorname{Maps}(M, S^1)$, $\operatorname{ad} L \cong M$ so the spaces $\Omega^i(\operatorname{ad} L)$ are actually the usual deRham vector spaces $\Omega^i(M)$. Moreover, the maps D_A turn out to be the same as the usual deRham maps! So in the $U(1)$ setting, we just recover the usual deRham complex.

Exercise 1.50
Check that $D_A = d$.

In this simpler setting, the Bianchi identity ($D_A F_A = 0$) becomes $d(da) = 0 \Rightarrow d^2 = 0$, which we already know to be true. The Yang Mills equation ($D_A^* F_A = 0$) becomes $d^* d = 0$, so in this setting, YM connections are exactly those which are harmonic.

Lemma 1.51
A U(1) Yang Mills connection is precisely a U(1) connection with harmonic curvature.

The upshot here is that Self Dual Yang Mills connections are U(1) connections with self-dual curvature, and Anti Self Dual Yang Mills connections are those with anti-self-dual curvature.

We've translated what it means to be Yang Mills, but what about action of the gauge group? There aren't that many harmonic forms; in particular, there are only as many as the number of de Rham cohomology classes. But in principle, there could be many connections with the same curvature. And these are the objects we actually care about. Recall that the Yang Mills moduli space is a space of connections, not curvatures, even though the condition is on curvatures.

So we must ask: how well do we understand the equivalence relation on connections? Formally, what is action $\mathcal{G} \curvearrowright \mathcal{A}$? A connection a maps to $a + g^{-1} dg$, so our question becomes what is $g^{-1} dg$? To answer this, we first simplify the topology of M.

Lemma 1.52
Let $H_1(M) = 0$. Then two connections ∇, ∇' differ by a U(1) gauge transformation iff they differ by an exact 1-form: $\nabla - \nabla' = i\, dh$ for some purely imaginary $h \in \Omega^0$.

Remark 1.53
In general, $\nabla - \nabla' \in \Omega^1(\operatorname{ad} L)$, but $\operatorname{ad} L$ is trivial for U(1), so $\nabla - \nabla' \in \Omega^1$.

Proof.

($\Leftarrow$) If $\nabla - \nabla'$ is an exact 1-form, then $\nabla - \nabla' = i\, dh$. Set $g = e^{ih}$. This is an S^1-valued function on M. Thus

$$g^{-1} dg = (e^{ih})^{-1} d(e^{ih}) = \frac{1}{e^{ih}} \cdot i\, dh \cdot e^{ih} = i\, dh.$$

$\Rightarrow$ If $\nabla - \nabla' = g^{-1} dg$ for some $g : M \to S^1$, to get the difference as an exact 1-form, we must first get a 1-form. Thus we must lift $g : M \to S^1$ to a map to $\mathbb{R}$:

$$\begin{tikzcd} & \mathbb{R} \arrow[d, "\exp"] \\ M \arrow[ur, "h=\log g"] \arrow[r, "g"] & S^1 \end{tikzcd}$$

By covering space theory, a lift of g exists if $g^*(H_1(M)) = 0$. By assumption, $H_1(M) = 0$, so g lifts to $h := \log g : M \to \mathbb{R}$.

Now set $g = e^{ih}$. Then $g^{-1}dg = i\,dh$, thus $\nabla - \nabla'$ is an exact one form.

-

□

Lemma 1.54

Let $H_1(M) = 0$. Then for each L, there exists a unique (up to gauge transformation) YM U(1) connection on L. The curvature of this L is given by the unique harmonic representative of $c_1(L)$.

Proof. We first show that every L admits a YM connection. Let ω be the (unique) harmonic representative of $c_1(L)$, and pick any U(1) connection ∇ on L. We know that ∇ has curvature $d\alpha$, where $[d\alpha] = [\omega]$, since Chern classes are well-defined. From ∇, we now build a connection whose curvature is literally ω, not just a 2-form in the same equivalence class as ω.

Since $\mathcal{A}$ is affine over $i\Omega^1(M)$, by choosing an appropriate $b \in \Omega^1(M;\mathbb{R})$, we can define another connection $\nabla' := \nabla + ib$ whose curvature is $d\alpha + i\,db = \omega$.

We now must show ∇' is unique up to U(1) gauge. Any other ∇ must have the same curvature by definition.

Thus

$$0 \equiv F_\alpha - F_{\alpha'} = d(\alpha - \alpha')$$

$$\equiv \nabla - \nabla' \text{ is a closed 1-form}$$

$$\equiv \nabla - \nabla' \text{ is an exact 1-form, since } H^1_{dR} = 0$$

$$\equiv \nabla - \nabla' = dh \text{ for some } h \in \Omega^0(M)$$

$$\equiv \nabla - \nabla' = g^{-1}dg \text{ for some } g : M \to S^1$$

So ∇ and ∇' must differ by a gauge transformation, making them the same element modulo gauge. □

The upshot here is that the Yang Mills moduli space in U(1) is a single point for each choice of L.

Lemma 1.55

Let $H_1(M) = 0$.

- If M is positive definite, for every L, the YM connection is SD.
- If M is negative definite, for every L, the YM connection is ASD.

Before we prove this lemma, we need to prove that harmonic 2-forms also decompose into self dual and anti self dual subspaces.

Proposition 1.56
The space of harmonic 2-forms decomposes as the direct sum

$$\mathcal{H}^2 = \mathcal{H}^2_+ \oplus \mathcal{H}^2_-,$$

where $\mathcal{H}^2_+ = \{\text{self-dual harmonic 2-forms}\}$ and $\mathcal{H}^2_- = \{\text{anti-self-dual harmonic 2-forms}\}$.

Proof. Recall $\Omega^2 = \Omega^2_+ \oplus \Omega^2_-$. Thus any $\omega \in \mathcal{H}^2$ splits as $\omega = \omega_+ + \omega_-$, but $\omega_\pm$ are not necessarily harmonic.

By harmonicity,

$$\begin{cases} d\omega = 0 \;\Rightarrow\; d\omega_+ + d\omega_- = 0 \\ d^*\omega = 0 \;\Rightarrow\; d^*\omega_+ + d^*\omega_- = 0. \end{cases}$$

By (anti) self-duality of $\omega_\pm$,

$$\begin{cases} \star d \star (\omega_+) = \omega_+ \\ \star d \star (\omega_-) = -\omega_-. \end{cases}$$

Therefore $d^*\omega_+ = \star d\omega_+$ and $d^*\omega_- = -\star d\omega_+$, thus

$$\star d\omega_+ - \star d\omega_- = 0$$

$$\star d\omega_+ = \star d\omega_- = 0$$

$$d\omega_+ = d\omega_- = 0.$$

So $\omega_\pm$ are themselves harmonic. ☐

We can now prove the lemma.

Proof of Lemma. If M is positive definite, Q_M is strictly positive. Therefore for any $\alpha \in \mathcal{H}^2(M)$,

$$Q_M(\alpha, \alpha) = \int_M \alpha \wedge \alpha = \|\alpha\|_{L^2} > 0,$$

so there are no ASD harmonic forms.

The same argument holds for negative definite. ☐

Armed with these results for U(1) connections, we can now try to upgrade these to results for SU(2) connections. In particular, sometimes an SU(2) connection will split as the sum of two U(1) connections.

Split Yang-Mills Connections

Let $E \to M$ be an SU(2) bundle which splits as $E = L \oplus L'$, where L and L' are U(1) bundles. If so, L' must be $\bar{L}$ by the matrix constraints for U(1) and SU(2).

So we'll analyse SU(2) connections by first restricting to split connections $\nabla = \nabla \oplus \bar{\nabla}$. The corresponding 1-form matrix for ∇ is

$$A = \begin{pmatrix} i\,da & 0 \\ 0 & -i\,da \end{pmatrix},$$

and the curvature also splits as

$$F_A = \begin{pmatrix} i\,db & 0 \\ 0 & -i\,db \end{pmatrix},$$

where $i\,db$ represents $c_1(L)$ and $-i\,db$ represents $c_1(\bar{L}) = -c_1(L)$.

Lemma 1.57

Let M^4 be positive definite with $H_1(M) = 0$. Take E an SU(2) bundle over M. For each splitting $E = L \oplus \bar{L}$, there is a unique (modulo gauge) split self-dual YM connection on E.

Proof (sketch).

1. Construct a self-dual U(1) YM connection ∇ on L as before.
2. Check that the conjugate connection $\bar{\nabla}$ is a self-dual connection on $\bar{L}$.
3. Their sum $\nabla \oplus \bar{\nabla}$ is thus a self-dual SU(2) connection on $L \oplus \bar{L}$.
4. We saw that ∇ is unique up to U(1) gauge. Check that any other split SU(2) connection $\nabla' \oplus \bar{\nabla}'$ is gauge equivalent to $\nabla \oplus \bar{\nabla}$ via some split gauge transformation.

□

Remark 1.58

If E splits, this splitting happens at the level of a reduction of the structure group. So the local splitting we observed before as a matrix change of basis for A must not only happen in a neighborhood U_α, but must be a simultaneous change of basis and transition maps for all matrices A on every neighborhood U_α.

Lecture 7 // Sept 15 // Outline of proof of Donaldson's theorem

Last time:
Let (M^4, g) be positive definite and simply connected. Fix a $k = 1$ SU(2) bundle E over M. Set

$$\mathcal{M} := \{\text{SD YM connections on } E\}/\mathcal{G}.$$

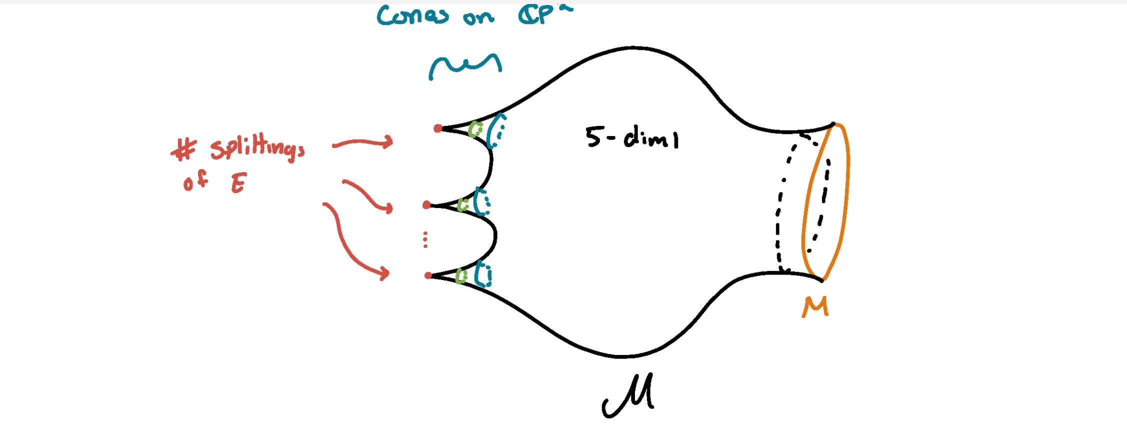
Each splitting $L \oplus \bar{L}$ of E has a unique (up to SU(2) gauge equivalence) split SDYM connection $\nabla \oplus \bar{\nabla}$ for which F_∇ is the unique harmonic representative of $c_1(L)$.

Today, we'll outline a proof of Donaldson's theorem by establishing qualitative features about $\mathcal{M}$.

Theorem (Donaldson (**??**))
If X^4 is $\pi_1 = 1$, smooth, closed 4-manifold and Q_X is definite, then Q_X is equivalent to $n(+1)$ or $n(-1)$.

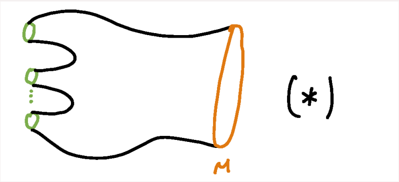
Claim 1.59.

1. *Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dimensional oriented manifold.*
2. *The number of these points = the # of splittings of E.*
3. *Each point has a neighborhood homeomorphic to cone on $\mathbb{CP}^2$; diffeomorphic away from point.*
4. *$\mathcal{M}$ is not compact, but it admits a compactification $\bar{\mathcal{M}}$ such that $\partial\bar{\mathcal{M}} \cong M$.*



We have a cobordism M to $\bigsqcup_{\substack{\#\text{splittings}\\ \text{of } E}} \mathbb{CP}^2$ obtained by removing the singular points. (∗)

It turns out that (∗) implies Donaldson's theorem. Before we discuss this, we'll briefly motivate why bordism is useful here.



Interlude: Bordism in General

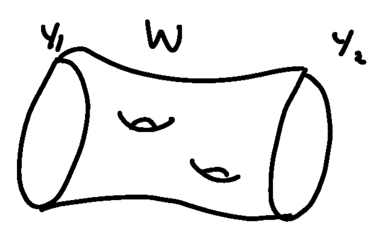
This discussion isn't strictly necessary, but it places $(*)$ in context.

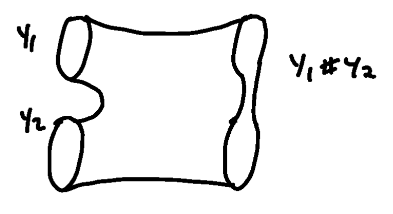
Definition 1.60

The n-th bordism group is

$$\Omega^n = \{\text{Set of all oriented smooth } n\text{-mfd.}\} / \sim$$

where $Y_1 \sim Y_2$ iff there exists W^{n+1} such that $\partial W = -Y_1 \sqcup Y_2$. If $Y_1 \sim Y_2$, they are called bordant. Ω^n is a group under disjoint union and/or connect sum.





This is a generalization of the question: is $M^n \cong \partial W^{n+1}$ for some W^{n+1}? It turns out that in one of the great algebraic topology triumphs of the 60's, the groups Ω^n are well understood. In particular, Ω^n/torsion is known for all $n \in \mathbb{N}$. Concretely, for $n = 3$, $\Omega^3 = 0$ since every 3-manifold bounds a 4-manifold. When $n = 4$, $\Omega^4 \cong \mathbb{Z}$, which is determined by signature.

Definition 1.61
The signature of a 4-manifold is $\sigma(M) = \sigma(Q_M) = \#{+}\text{eigenvalues} - \#{-}\text{eigenvalues}$.

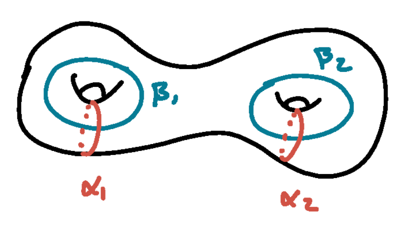
Theorem 1.62
In dimension 4, signature is a complete bordism invariant.

This means that two 4-manifolds are cobordant if and only if they have the same signature. In particular, any M is cobordant to either $\#_{\sigma(M)}\mathbb{C}\mathrm{P}^2$, if $\sigma(M) \geq 0$, or $\#_{-\sigma(M)}\overline{\mathbb{C}\mathrm{P}^2}$, if $\sigma(M) < 0$.

For our purposes, we'll only need the claim that σ is a bordism invariant.

Proposition 1.63
If M_1 and M_2 are bordant, then $\sigma(M_1) = \sigma(M_2)$.

To prove this proposition, we'll need the following lemma, commonly referred to as the "half lives, half dies" theorem.



To motivate it, consider $Z_g = \partial H_g$. Think of Z_g as a stand in for M and H_g a stand in for W. The middle dimension of the (co)homology of Z is H_1, so we look at the map induced by the inclusion $\iota : Z_g \to H_g$.

$$\iota_* : H_1(Z_g) \longrightarrow H_1(H_g)$$

$$\ker \iota_* \cong \operatorname{span}\{\alpha_1, \alpha_2\}$$

$$\operatorname{im} \iota_* \cong \operatorname{span}\{\iota_*(\beta_1), \iota_*(\beta_2)\}$$

Lemma 1.64 (Half lives, half dies)
For W a smooth oriented $2n+1$ dimensional manifold with boundary $\partial W \cong M$, $\ker(\iota_*) \subset H_n(M;\mathbb{R})$ is a half-dimensional Lagrangian subspace with respect to the intersection form $Q : H^n(M;\mathbb{R}) \times H^n(M;\mathbb{R}) \to \mathbb{Z}$, where $\iota_* : H_n(M;\mathbb{R}) \to H_n(W;\mathbb{R})$ is the map on homology induced by the inclusion $\iota : M \cong \partial W \hookrightarrow W$.

Proof of Lemma. First recall that by Poincaré duality, $H_n(M;\mathbb{R}) \cong H^n(M;\mathbb{R})$.

Further, from linear algebra, if $T : A \to B$ is a linear map, the dual map $T^* : B^* \to A^*$ is such that $\dim \operatorname{im} T = \dim \operatorname{im} T^*$:

$$\dim \operatorname{im} T = \dim(A) - \dim(\ker T) = \dim(A/\ker T) = \dim(\operatorname{Ann}(\ker T)) = \dim \operatorname{im} T^*$$

$$(\operatorname{Ann}(\ker T) \text{ is the subspace of } A^* \text{ that annihilates } T)$$

Thus the maps

$$\iota^* : H^n(W;\mathbb{R}) \longrightarrow H^n(M;\mathbb{R}), \qquad \iota_* : H_n(M;\mathbb{R}) \longrightarrow H_n(W;\mathbb{R})$$

are dual in the linear algebra sense, so $\dim \operatorname{im} \iota^* = \dim \operatorname{im} \iota_*$.

Again by Poincaré duality, $\dim \operatorname{im} \iota^* = \dim \ker \iota_*$.

Thus

$$\dim \ker \iota_* + \dim \operatorname{im} \iota_* = \dim H_n(M;\mathbb{R})$$

$$\dim \operatorname{im} \iota^* + \dim \operatorname{im} \iota^*$$

$$\Rightarrow \dim \operatorname{im} \iota^* = \tfrac{1}{2} \dim H_n(M;\mathbb{R}) = \tfrac{1}{2} \dim H^n(M;\mathbb{R})$$

$$\Rightarrow \dim \ker \iota_* = \tfrac{1}{2} \dim H_n(M;\mathbb{R})$$

Now, take $\alpha, \beta \in H^n(W;\mathbb{R})$, so $\iota^*(\alpha), \iota^*(\beta) \in H^n(M;\mathbb{R})$.

Then

$$Q_M\big(\iota^*(\alpha), \iota^*(\beta)\big) = \int_M \iota^*(\alpha) \wedge \iota^*(\beta) = \int_M \iota^*(\alpha \wedge \beta) = \int_{\partial W} \iota_*(\alpha \wedge \beta)$$

By Stokes' theorem,

$$\int_{\partial W} \iota_*(\alpha \wedge \beta) = \int_W d(\alpha \wedge \beta) = 0.$$

Thus $\operatorname{im}(\iota^*)$ is Lagrangian with respect to Q_M. □

Remark 1.65

We may avoid using Stokes' theorem by translating to algebraic topology and cupping against the fundamental class. Thus half lives half dies is true when M isn't smoothable.

Having proven half lives half dies, we may now prove the original proposition.

Proposition (??)

If M_1 and M_2 are bordant, then $\sigma(M_1) = \sigma(M_2)$.

Proof.

1. By setting $M = -M_1 \# M_2$, we know that $M \cong \partial(W^5)$. So it is enough to show that $\sigma(M) = 0$.
2. Consider the long exact sequence in homology and the Poincaré-Lefschetz isomorphism. By naturality, the following diagram commutes:

$$\begin{tikzcd}
\cdots \arrow[r] & H^2(W;\mathbb{R}) \arrow[r, "\iota^*"] \arrow[d, "PD"', "\cong"] & H^2(M;\mathbb{R}) \arrow[r] \arrow[d, "PD"', "\cong"] & H^3(W,M;\mathbb{R}) \arrow[r] \arrow[d, "PD"', "\cong"] & \cdots \\
\cdots \arrow[r] & H_3(W,M;\mathbb{R}) \arrow[r] & H_2(M;\mathbb{R}) \arrow[r, "\iota_*"] & H_2(W;\mathbb{R}) \arrow[r] & \cdots
\end{tikzcd}$$

   Note that $\operatorname{im}\iota^* \cong \ker \iota_*$ under PD.

3. By Half lives, half dies,

$$\begin{cases} \dim \operatorname{im}\iota^* = \frac{1}{2}\dim H^2(M;\mathbb{R}) \\ \dim \ker \iota_* = \frac{1}{2}\dim H_2(M;\mathbb{R}) \end{cases}$$

   and Q_M vanishes on $\operatorname{im}\iota^*$ (e.g., $\operatorname{im}\iota^* \subset H^2(M;\mathbb{R})$ is Lagrangian).

   Thus there is a half-dimensional subspace on which the intersection form vanishes.

4. This implies $\sigma(M) = 0$: consider the diagonalization of Q_M over $\mathbb{R}$: $Q_M \cong \begin{pmatrix} +I & 0 \\ 0 & -I \end{pmatrix}$.

$$\begin{pmatrix} +I & 0 \\ 0 & -I \end{pmatrix} V = V_+ \oplus V_-$$

   This splits V into $V_+ \oplus V_-$, where Q_M is $\pm$-definite on $V_\pm$.

   Let $a = \dim V_+$ and $b = \dim V_-$. Then $a + b = \dim V_+ + \dim V_- = \dim V = 2k$.

   Let $U = \operatorname{im}\iota^*$. Since Q_M vanishes on U, $U \cap V_+ = \{0\}$ and $U \cap V_- = \{0\}$.

   Thus

$$\dim(U + V_+) = \dim(U) + \dim(V_+) = k + a \leq 2k$$

$$\dim(U + V_-) = \dim(U) + \dim(V_-) = k + b \leq 2k$$

$$\Rightarrow a \leq k,\ b \leq k. \text{ Since } a + b = 2k,\ a = b = k \Rightarrow \sigma(M) = a - b = 0.$$

□

Back to Donaldson

Let's return to Donaldson and take stock of what we know:

> **Theorem** (Donaldson (**??**))
> If X^4 is $\pi_1 = 1$, smooth, closed 4-manifold and Q_X is definite, then Q_X is equivalent to $n(+1)$ or $n(-1)$.

The previous discussion of bordism has given us the following three facts, from which Donaldson's theorem will follow.

(i) $\sigma(M) = \mathrm{rk}(Q_M)$ (since M is positive definite).

(ii) $\sigma(M) = \#$ splittings of E.

Since

  (a) σ bordism invariant (Interlude)

  (b) M bordant to $\displaystyle\coprod_{\substack{\#\text{splittings}\\ \text{of } E}} \mathbb{CP}^2$ $(*)$

(iii) $\#$ splittings of $E = \frac{1}{2}\#x \in H^2$ such that $\langle x, x\rangle = 1$.

This follows from two claims about vector bundles over X^4:

  (a) SU(2) bundles are determined by $c_2 \in H^4$ (true only in our setting)

  (b) U(1) bundles are determined by $c_1 \in H^2$ (true in general)

Via the curvature form or by formal properties of Chern classes, to say $E = L \oplus \bar{L}$ implies that $c_2(E) = c_1(L) \cup c_1(\bar{L}) = -c_1(L) \cup c_1(L)$.

Similarly, if $c_2(E) = -x^2$ for $x \in H^2$, the above calculation implies that $E = L \oplus \bar{L}$. Note here that different choices of x give different L's in the splitting.

Therefore, if $k = c_2(E) = 1$, we must search over $x \in H^2$ for which $x^2 = 1$.

Combining (i), (ii), and (iii), we can see that

$$\sigma(M) = \mathrm{rk}(Q_M) = \#\text{splittings of } E = \tfrac{1}{2}\#x \in H^2 \text{ s.t. } x^2 = 1.$$

Heuristically, this equality says that Q_M has "many" x with $\langle x, x\rangle = 1$, where "many" is exactly twice the number of splittings.

To prove Donaldson's theorem, it remains to show that Q_M is equivalent to $n(+1)$ or $n(-1)$, i.e., $\pm n = \mathrm{rk}(Q_M) = \frac{1}{2}\#x \in H^2$ such that $x^2 = 1$.

> **Lemma 1.66**
> Let $Q : V \times V \to \mathbb{Z}$ be a positive-definite symmetric unimodular bilinear pairing. Then $\#x$ such that $Q(x, x) = 1 \leq 2\,\mathrm{rk}\, Q$, with equality iff Q is equivalent to I over $\mathbb{Z}$.

Assuming the lemma, since $\mathrm{rk}(Q_M) = \frac{1}{2}\#x \in H^2$ such that $x^2 = 1$, where x^2 is shorthand for $Q(x,x)$, we get that

$$\#x \text{ s.t. } Q(x,x) = 1 \leq 2 \cdot \tfrac{1}{2}\#x \in H^2 \text{ s.t. } Q(x,x) = 1$$

$$\#x \text{ s.t. } Q(x,x) = 1 \leq \#x \in H^2 \text{ s.t. } Q(x,x) = 1$$

So clearly we must have equality, so the lemma implies that Q is equivalent to I over $\mathbb{Z}$, thus $Q \cong \pm(n)$. This proves Donaldson's theorem.

We now prove the lemma.

Proof of Lemma. First note that if $Q = I$, there is clearly $2 \cdot \mathrm{rk}\, Q$ of such elements of H^2, since the standard basis vectors e_i have self pairing 1, as well as $-e_i$.

For a general Q, we'll apply the Gram-Schmidt process over $\mathbb{Z}$. Let x be such that $Q(x,x) = 1$. We claim that $V = \langle x \rangle \oplus \langle x \rangle^\perp$. Indeed, since any $z \in V$ decomposes as

$$z = \underbrace{Q(x,x)x}_{\in\langle x \rangle} + \underbrace{(z - Q(x,z)x)}_{\in\langle x \rangle^\perp}$$

we know that $V = \langle x \rangle + \langle x \rangle^\perp$; since $\langle x \rangle \cap \langle x \rangle^\perp = \{0\}$, this decomposition is orthogonal.

Thus $Q : \langle x \rangle \oplus \langle x \rangle^\perp \times \langle x \rangle \oplus \langle x \rangle^\perp \to \mathbb{Z}$, so Q splits as a block matrix

$$Q = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & * \end{array}\right) \begin{array}{l} \big\} \ \langle x \rangle \\ \big\} \ \langle x \rangle^\perp \end{array}$$

Now, notice the following:

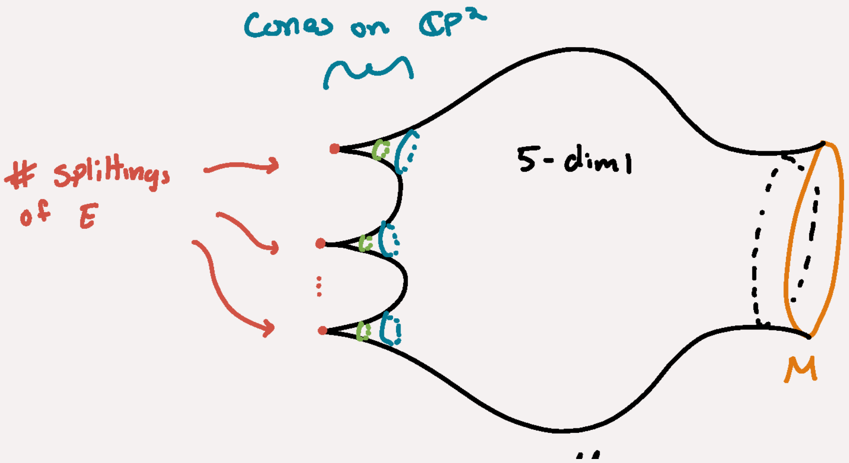
1. Q is still positive definite on $x^\perp$.
2. Any $z \in V$ such that $Q(z,z) = 1$ is either $\pm x$ or lies entirely in $\langle x \rangle^\perp$: if $z = cx + w$ for $w \in x^\perp$, $Q(z,z) = c^2 + \langle w, w \rangle \neq 1$.
3. The rank of $Q|_{\langle x \rangle^\perp}$ is $\mathrm{rank}(Q) - 1$.

Thus we may proceed by induction on $Q|_{\langle x \rangle^\perp} : \langle x \rangle^\perp \times \langle x \rangle^\perp \to \mathbb{Z}$. If $\langle x \rangle^\perp$ contains no elements for which $x^2 = 1$, the process concludes, and the number of elements for which $x^2 = 1$ is necessarily less than $2 \cdot \mathrm{rk}(Q)$. Otherwise, Q can be completely diagonalized, so Q is equivalent to I. □

Lecture 8 // Sept 17 // Singularities in $\mathcal{M}$ & Holonomy

Last Time: Let (M^4, g) be positive definite and simply connected. Fix a $k = 1$ SU(2) bundle E over M. Consider $\mathcal{M} = \mathcal{A}/\mathcal{G}$ the SDYM moduli space. We know the following qualitative features of $\mathcal{M}$:

1. Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dimensional oriented manifold.
2. The number of these points = the number of splittings of E.
3. Each point has a neighborhood homeomorphic to the cone on $\mathbb{CP}^2$; diffeomorphic away from the point.
4. $\mathcal{M}$ is not compact, but it admits a compactification $\overline{\mathcal{M}}$ such that $\partial\overline{\mathcal{M}} \cong M$.



After some linear algebra, these properties prove Donaldson's theorem: since σ is a bordism invariant,

$$\underbrace{\mathrm{rk}\, Q_M}_{\text{since definite}} = \sigma(M) = \#\text{ splittings of } E = \frac{1}{2}\#\{x \in H_2 \text{ with } x^2 = 1\}.$$

Today, we'll unpack these points: why is $\mathcal{M}$ singular, and why do neighborhoods look like the cone on $\mathbb{CP}^2$?

Singularities in $\mathcal{M}$

Recall that $\mathcal{M} = \{\text{SDYM conn.}\}/\mathcal{G}$. $\{\text{SDYM conn.}\} \subseteq \mathcal{A}$, which is a smooth manifold, on which $\mathcal{G}$ acts. However, $\mathcal{G}$ may have stabilizer subgroups. Thus the singularities arise from the failure of $\mathcal{G}$ to act freely on $\mathcal{A}$.

Remark 1.67

Establishing $\{\text{SDYM conn.}\}$ is a smooth manifold is fairly standard: it is the zero locus of a map from an ∞-dimensional manifold to somewhere else; verify that 0 is a regular value, apply Sard-Smale.

Definition 1.68

For $\nabla \in \mathcal{A}$, $\mathrm{Stab}(\nabla) = \{g \in \mathcal{G} \mid g^*(\nabla) = \nabla\}$,

$$A \longrightarrow g^{-1}Ag + g^{-1}dg.$$

What can $\mathrm{Stab}(\nabla)$ be? First note that for every ∇, $\pm\mathrm{Id} \in \mathrm{Stab}(\nabla)$. Since $\pm\mathrm{Id}$ is sent to itself under conjugation, $\pm\mathrm{Id}$ is a well-defined section of $\mathrm{Aut}(E)$. These are not interesting gauge transformations, so it's sometimes more convenient to take $\mathcal{G}/\pm\mathrm{Id} \curvearrowright \mathcal{A}$ instead.

What else is possible? Let's examine an example. Connections with nontrivial stabilizers are related to the number of splittings. For a line bundle L, consider the constant gauge transformation given by multiplication by a fixed element of S^1, $e^{i\theta} \in S^1$. For $E = L \oplus \bar{L}$, consider now

$$g = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}.$$

One can check this is a well-defined gauge transformation on E. If ∇ is split, e.g., $A = \begin{pmatrix} ia & 0 \\ 0 & -ia \end{pmatrix}$, easy to check that $g^*(\nabla) = \nabla$. So if E is split and ∇ is a split connection, considered as a subgroup of $\mathrm{SU}(2)$, $\mathrm{Stab}(\nabla)$ is at least as big as S^1.

Remark 1.69

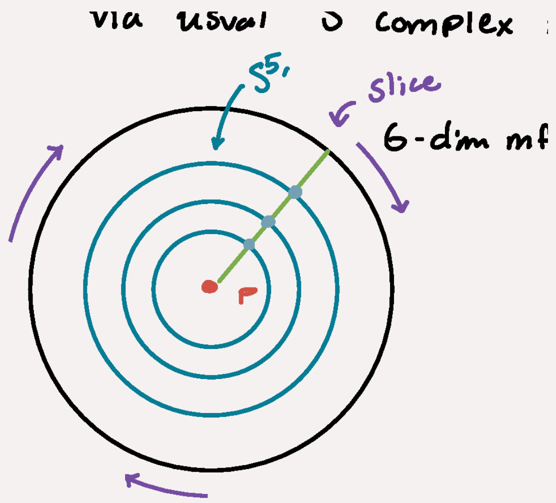
In this observation, we play fast and loose with the difference between a local definition of a gauge transformation and its definition globally. The correct statement is given in the following lemma, and the tension between global & local will be resolved as we introduce the tools to prove it.

Lemma 1.70

For an $\mathrm{SU}(2)$ bundle with $k \neq 0$, a connection has stabilizer bigger than $\pm\mathrm{Id}$ iff it is split, and $\mathrm{Stab}(\nabla) \cong S^1$.

The proof isn't hard, but requires more technical language. But this lemma gives why the singularities of $\mathcal{M}$ are what they are. We showed previously that each splitting of E supports a unique (up to $\mathcal{G}$) split SDYM connection. Each singular point is the unique SDYM connection for a splitting. Thus the number of singular points is the number of splittings. This gives (2).

We now ask why local neighborhoods are cones on $\mathbb{CP}^2$. Near a singular point, $\mathcal{G}/\pm I$ acts freely. The approximate idea is that we first quotient by $\pm I$ to get a 6-dimensional manifold with a $S^1/\pm I$ action which acts freely except at one point. The local model for this is $S^1 \curvearrowright \mathbb{R}^6 \cong \mathbb{C}^3$ via usual S^1 complex multiplication. $\mathbb{C}^3/S^1 \cong$ cone on $\mathbb{CP}^2$ with one singular point, since on each S^5 slice, we have $S^5/S^1 \cong \mathbb{CP}^2$ (via definition $\mathbb{CP}^n := S^{2n+1}/S^1$).



Note 1.71

The $\pm I \subset \mathcal{G}$ and S^1 actions are more entertwined in $\mathcal{G}$ than this; we can't actually quotient by gauge in this "partial" way. But it gives an approximate idea.

Question 1.72. *Why is* $\mathcal{M}$ 5*-dimensional?*

This is where we use that $k = 1$. The dimension is from the Atiyah-Singer index formula applied to a Fredholm operator whose zero locus are the SDYM connections. When $k > 1$, we get more generally that the singular neighborhood is $\cong$ cone on $\mathbb{CP}^n$.

We now must introduce more differential geometry.

Interlude: Parallel Transport

Let E be a vector bundle over M.

Definition 1.73

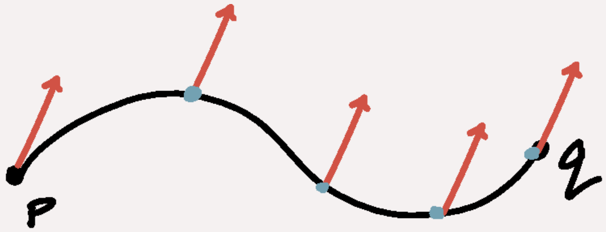
Let ∇ be a connection on E, $\gamma : [0, 1] \to M$ a path. A section s defined along γ is parallel transported by ∇ along γ if $\nabla_{\gamma'(t)} s = 0$ for all t.

Lemma 1.74

Let $\gamma : [0, 1] \to M$ be a path from p to q. If we fix any $v \in E_p$, then there exists a unique section s defined along γ such that s extends v and s is parallel transported by ∇ along γ.

Proof idea. Using the existence of solutions to ODEs, we can construct s and show that $(d + A)s = 0$. ☐

To better see the point of parallel transport, we can observe the effect of parallel transport on a frame of E_p, i.e., a basis for E_p, and parallel transport that basis along γ. Parallel transport preserves orthonormality, so this gives you a particular trivialization of E along γ for which the basis elements are covariantly constant: A vanishes in the path direction γ'. In essence, the covariant derivative specifies a "preferred frame" along each path, which we can find by choosing a frame at one end and parallel transporting it along the path.



What else can parallel transport do? For each path γ from p to q, we can build a linear map from E_p to E_q which sends $v \in E_p$ to the endpoint of the parallel transport of v along γ. Call this map $\mathrm{hol}_{p,q}(\gamma) : E_p \to E_q$. Since $\mathrm{hol}_{p,q}(\gamma) \circ \mathrm{hol}_{q,p}(\bar{\gamma}) = \mathrm{id}$, this map is an isomorphism.

Now, take $q = p$, so γ is a based loop in M. Then $\mathrm{hol}_p = \mathrm{hol}_{p,p}$ gives a map $E_p \to E_p$, so defines an element $\mathrm{hol}_p(\gamma) \in \mathrm{GL}(E_p)$.

Definition 1.75

Holonomy is a map $\mathrm{hol}_p : \{\text{loops based at } p\} \to \mathrm{GL}(E_p)$.

Properties of Holonomy:

1. If E is a G-bundle and ∇ is a G connection, then $\mathrm{hol}_p(\gamma)$ is $G \subseteq \mathrm{GL}(E_p)$ valued.
2. If $F_\nabla = 0$, the holonomy only depends on the based smooth homotopy class of γ. I.e., hol_p descends to a map
$$\mathrm{hol}_p : \pi_1(M) \to \mathrm{GL}(E_p) \quad (\text{or } G \text{ for } G).$$
For $G = \mathrm{SU}(2)$, $\mathrm{hol}_p : \pi_1(M) \to \mathrm{SU}(2)$ is a $\mathrm{SU}(2)$ representation of $\pi_1(M)$.
3. If g is a gauge transformation,
$$\mathrm{hol}_{p,q}^{g^*(\nabla)}(\gamma) = g_q^{-1} \circ \mathrm{hol}_{p,q}^{\nabla}(\gamma) \circ g_p.$$

To illustrate holonomy, consider a trivial line bundle $L = M \times \mathbb{C}$ over M and a $U(1)$ connection on L given by $\nabla = d + ia$, $a \in \Omega^1(M; \mathbb{R})$. Thus a section here is just a $\mathbb{C}$-valued function. For a path γ, what does it mean for $\nabla_{\gamma'(t)} s = 0$? Given a based loop γ and a section defined along γ, what does it mean for s to be parallel transported along γ? Since we're working in a line bundle, a section is just a $\mathbb{C}$-valued function, and for $\nabla_{\gamma'(t)} s = 0$, the covariant derivative of this function must vanish along γ. Explicitly, we get

$$\nabla_{\gamma'(t)} s = 0, \qquad (ds)(\gamma'(t)) + ia(\gamma'(t))s = 0, \qquad \frac{d}{dt}(s \circ \gamma) + ia(\gamma'(t))(s \circ \gamma(t)) = 0.$$

But this is just a differential equation, which we can solve as usual:

$$\frac{d}{dt}(s \circ \gamma) = -ia(\gamma'(t))(s \circ \gamma(t)) \;\Rightarrow\; (s \circ \gamma) = \left(e^{-i \int_0^t a(\gamma'(h))\, dh}\right) s(\gamma(0)).$$

If we conflate s and $s \circ \gamma$, we get that $\mathrm{hol}_{p,q} = e^{-i\int_\gamma a} \in S^1$.

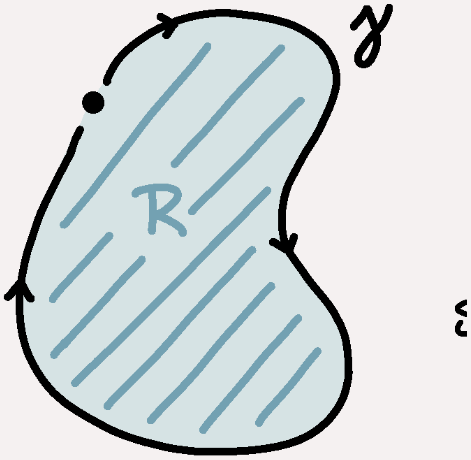
Intuitively, we're supposed to think that solving $(d + A)s = 0$ gives $s(t) = e^{-\int A} s(0)$, so if A is Lie algebra valued, $-\int A$ is Lie algebra valued, so $e^{-\int A}$ is Lie group valued. In reality, this is false – there's an issue with commutativity when $G \neq U(1)$.

However we see directly here that $\mathrm{hol}(\gamma)$ need not be the identity: if $\int_\gamma a \neq 2\pi n$, $e^{-i\int_\gamma a} \neq 1$.

In fact we see a nice connection with property #3. Indeed, suppose that γ bounds a region R. Then Stokes' theorem says that

$$\int_\gamma a = \int_{\partial R} a = \int_R da,$$

where da is the curvature of ∇.



So the heuristic is that holonomy picks up the curvature of a connection.

This demonstrates, in the $U(1)$ case, assertion #3: if γ is varied through a region where $F_\nabla = 0$, then holonomy is not changed. However, note that this analysis applies to single coordinate charts; it is purely local. If the bundle is nontrivial, it is not true that $F_A = 0$ implies that $\mathrm{hol}(\gamma) = \mathrm{id}$. Furthermore, it makes no sense to treat the differential equation like it's globally defined, since a isn't. But it is true that holonomy is constant in its homotopy class.

Next time: We'll use parallel transport to discuss the lemma on the exact behavior of stabilizers of connections.

Lecture 9 // Sept 22 // Singularities in $\mathcal{M}$, part 2

Last time:

We saw that connections whose stabilizer is larger than $\pm\mathrm{Id}$ lead to singular points of $\mathcal{M}$, and this does happen.

If E is split and ∇ is split, then $\mathrm{Stab}(\nabla)$ is at least as large as S^1. Precisely, we had the following lemma:

$$\text{If } E \text{ is a } k \neq 0 \text{ SU}(2) \text{ bundle, then } \mathrm{Stab}(\nabla) \text{ is larger than } \pm I$$

$$\text{iff } \nabla \text{ is split, in which case } \mathrm{Stab}(\nabla) \cong S^1.$$

Towards proving this lemma, we discussed parallel transport and holonomy.

The general question is what are the possibilities of $\mathrm{Stab}(\nabla)$? A priori, we know that $\mathrm{Stab}(\nabla) \subseteq \mathcal{G}$, and this fact actually restricts what $\mathrm{Stab}(\nabla)$ must be as follows.

Lemma 1.76

$g \in \mathrm{Stab}(\nabla)$ iff $D_\nabla g = 0$, where $D_\nabla : \mathrm{End}(E) \to \mathrm{End}(T^*M \otimes E)$, i.e., g is parallel transported by D_∇ along every γ.

Proof. $D_\nabla g$ is characterized by the Leibniz rule:

$$\nabla(gs) = (D_\nabla g)s + g(\nabla s)$$

After moving terms, we get

$$g^{-1}\nabla(gs) = g^{-1}(D_\nabla g)s + \nabla s$$

By definition, $g^*(\nabla)s = g^{-1}\nabla(gs)$.

Thus

$$g \in \mathrm{Stab}(\nabla) \text{ iff } g^*(\nabla) = \nabla$$

$$\Leftrightarrow \nabla(s) = g^{-1}(D_\nabla g)s + \nabla s \quad \forall s$$

$$\Leftrightarrow g^{-1}(D_\nabla g)s = 0 \quad \forall s$$

$$\Leftrightarrow D_\nabla g = 0$$

□

Moreover, we know that the parallel transport is unique. So we can also conclude the following.

Lemma 1.77

If $g \in \mathrm{Stab}(\nabla)$, then g is determined by its value at a single point.

Remark 1.78

It's not true that any g can be transported: g need not be globally defined! But we know $g \in \mathrm{Stab}(\nabla)$, so a priori, it is a global object, so parallel transport makes sense.

Equivalently, we can write this as for any p, the restriction map $\mathcal{G} \to \mathrm{Aut}(E_p)$ is injective on $\mathrm{Stab}(\nabla)$,

$$g \longmapsto g_p.$$

Thus (via restriction) $\mathrm{Stab}(\nabla)$ can be considered as a subgroup of $\mathrm{SU}(2)$. In fact, $\mathrm{Stab}(\nabla)$ has a nice connection with the holonomy of ∇.

Recall that for a fixed $p \in M$, the holonomy gives a map

$$\mathrm{hol}_p : \{\text{loops based at } p\} \longrightarrow \mathrm{Aut}(E_p) = \mathrm{SU}(2)$$

$$\gamma \longmapsto \mathrm{hol}_p^{\nabla}(\gamma)$$

Very explicitly, under hol_p, each γ gets sent to a specific linear transformation E_p to E_p, e.g., a matrix transformation of the fibre, e.g., an element of $\mathrm{SU}(2)$. If γ is varied to another γ', we get another matrix transformation.

By considering every γ', the set of corresponding matrices is $\mathrm{im}(\mathrm{hol}_p)$, and since $\mathrm{hol}_p(\gamma_2 \circ \gamma_1) = \mathrm{hol}_p(\gamma_2) \circ \mathrm{hol}_p(\gamma_1)$, $\mathrm{im}(\mathrm{hol}_p)$ forms a subgroup of $\mathrm{SU}(2)$.

Lemma 1.79

The restriction of $\mathrm{Stab}(\nabla)$ to a point p is the centralizer of $\mathrm{im}\,\mathrm{hol}_p$.

Proof. ($\subseteq$): Take $g \in \mathrm{Stab}(\nabla)$. We want to show that g_p commutes with $\mathrm{hol}_p(\gamma)$ for every loop γ. We know that $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$.

Since $g \in \mathrm{Stab}(\nabla)$, we know $g^*(\nabla) = \nabla$, so $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = \mathrm{hol}_p^{\nabla}(\gamma)$.

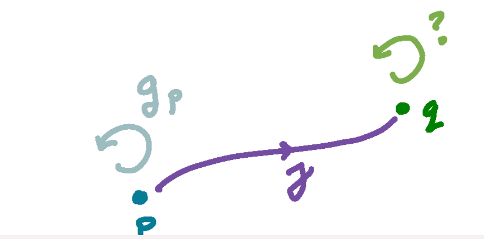
Thus $\mathrm{hol}_p^{\nabla}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$ as desired. ☐

($\supseteq$): Take $g_p \in \mathrm{Aut}(E_p)$. Further assume that $g_p \in Z(\mathrm{im}(\mathrm{hol}_p))$.

We want to show that there exists some $g \in \mathrm{Stab}(\nabla)$ such that the restriction of g to E_p is g_p.

Since $\mathrm{Stab}(\nabla) = \{g \mid D_\nabla g = 0\}$, to define g, we should try to extend g_p via parallel transport by D_∇.

For any other point q, let γ be a path from p to q.



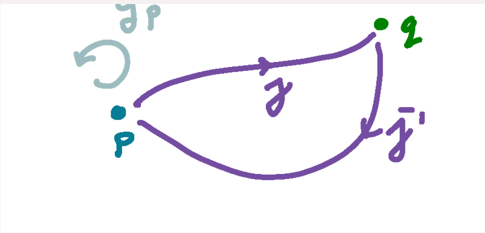
Define g_γ such that

① g_p is extended along γ such that $(D_\nabla)_\gamma g = 0$

② $g_q = \mathrm{hol}_{p,q}(\nabla) \circ g_p \circ \mathrm{hol}_{q,p}(\bar{\gamma})$

We can do this along γ since parallel transport exists. For any path, we can cook up a g_γ such that $D_\nabla(g_\gamma)$ vanishes along γ; this follows by existence of solutions to ODEs.

However a priori, we don't know that this extension along γ gives a well-defined global g. Take γ' another path from p to q, and consider g' defined by γ'.



Then g_q and g'_q are defined as

$$g_q = \mathrm{hol}(\gamma) \circ g_p \circ \mathrm{hol}(\bar{\gamma})$$

$$g'_q = \mathrm{hol}(\gamma') \circ g_p \circ \mathrm{hol}(\bar{\gamma}').$$

If g_q is well defined, $g_q^{-1} = (g'_q)^{-1}$ for any other g'_q, i.e.,

$$g_q(g'_q)^{-1} = \mathrm{id}.$$

Indeed,

$$g_q(g'_q)^{-1} = \mathrm{hol}(\gamma) \circ g_p \circ \mathrm{hol}(\bar{\gamma}) \left(\mathrm{hol}(\gamma') \circ g_p \circ \mathrm{hol}(\bar{\gamma}')\right)^{-1} = \mathrm{id} \qquad \square$$

Let's take stock so far: we know that Stab(∇) is determined by its restriction to a point, which is the centralizer of the holonomy at that point, viewed as a subgroup of SU(2).

So our question now becomes what are the subgroups of SU(2)?

Fact 1.80

They are as follows:

Subgroups of SU(2)	Centralizer = Stab(∇)
I	SU(2)
$-I$	SU(2)
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
Non-abelian	$\pm I$
SU(2)	$\pm I$

This means that the only possibilities for Stab(∇) are ±Id, S^1, or SU(2) itself. We are now in the position to prove the lemma.

Lemma 1.81

If E is a $k \neq 0$ SU(2) bundle, then Stab(∇) is larger than $\pm I$ iff ∇ is split, in which case Stab(∇) $\cong S^1$.

Proof. We first show that either Stab(∇) $\cong S^1$ or ±Id. From there, we'll show the former happens iff ∇ is split.

We just saw that there are only three possibilities for Stab(∇), so we need only to rule out SU(2).

If ∇ is a connection on an $k \neq 0$ bundle, then F_∇ cannot be identically 0 (if it was, $c_2 = k = 0$).

Thus holonomy can't be nontrivial, since the holonomy picks up curvature enclosed by a loop. If this path is varied, hol will vary continuously, so im(hol_p) can't be contained in a discrete subgroup. Thus the centralizer cannot be $\pm I$, so the subgroup can't be SU(2).

Now, suppose Stab(∇) = S^1. We just saw that Stab(∇) = S^1 implies that at a point $p \in M$, the element $g_p \in \mathrm{SU}(2)$ must actually be contained in $U(1) \subset \mathrm{SU}(2)$.

Therefore

$$g_p = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix},$$

which induces a splitting of the fibre over p by considering the eigenspaces.

For any other $q \in M$, g_q is obtained from g_p via conjugation with $\mathrm{hol}_{p,q}(\gamma)$, so the eigenspaces of g_q provide a splitting of E_q compatible with that of E_p. Since $\mathrm{Stab}(\nabla)$ is reconstructed from g_p by parallel transport, the splitting of E_p induces a splitting of E.

To Do: Insert proof of other direction. □ □

Having proved this lemma, we can conclude our discussion of why the number of singular points in $\mathcal{M}$ is the number of splittings of E.

We'll now begin a discussion of the other major qualitative feature $\mathcal{M}$: that $\partial\bar{\mathcal{M}} = M$. To do so, we'll first discuss instantons ((A)SD Yang-Mills connections) on S^4, which we'll do by identifying S^4 with $\mathbb{HP}^1$.

Interlude: Review of Quaternions & Some Useful Isomorphisms from Algebra

Definition 1.82

A quaternion is an expression of the form $q = x + yi + zj + wk$ where $x, y, z, w \in \mathbb{R}$ and $1, i, j, k$ are basis elements such that

$$1 \cdot \alpha = \alpha \text{ for } \alpha = i, j, k, \qquad i^2 = j^2 = k^2 = ijk = -1$$

$$i \cdot j = k, \quad j \cdot k = i, \quad k \cdot i = j, \qquad \text{and} \qquad \alpha \cdot x = x \cdot \alpha \text{ for } \alpha = i, j, k \text{ and } x \in \mathbb{R}.$$

The space of quaternions is denoted $\mathbb{H}$.

Remark 1.83

$\mathbb{H}$ admits the structure of a division ring. However, multiplication fails to be commutative, so $\mathbb{H}$ is not a field: if $q_1 = i$ and $q_2 = j$, $q_1q_2 = ij \neq ji = q_2q_1$.

Note that $\mathbb{H} \cong \mathbb{C}^2 \cong \mathbb{R}^4$ by the following isomorphisms. First rewrite a quaternion as

$$x + yi + zj + wk = (x + yi) + (zj + wk) = (x + yi) + (zj + wij) = (x + yi) + (z + wi)j.$$

Then the isomorphisms are given by the maps

$$\mathbb{H} \longrightarrow \mathbb{C}^2 \longrightarrow \mathbb{R}^4$$

$$(x + iy) + (z + iw)j \longmapsto (x + iy,\ z + iw) \longmapsto (x, y, z, w).$$

The quaternions can also be identified by a specific subgroup of the space of 2×2 complex matrices.

Definition 1.84

The Pauli matrices are a set of three 2×2 matrices $\{\sigma_1, \sigma_2, \sigma_3\} \subset \mathrm{Mat}_{2\times 2}(\mathbb{C})$ which are traceless, Hermitian, unitary, and involutive:

$$\sigma_1 := \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \qquad \sigma_2 := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \qquad \sigma_3 := \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

By sending $i \mapsto \sigma_1$, $j \mapsto \sigma_2$, and $k \mapsto \sigma_3$, any quaternion $q = x + yi + zj + wk$ can be rewritten as the linear combination

$$\begin{aligned} x\,\mathrm{Id} + y\sigma_1 + z\sigma_2 + w\sigma_3 &= \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} + \begin{pmatrix} yi & 0 \\ 0 & -yi \end{pmatrix} + \begin{pmatrix} 0 & z \\ -z & 0 \end{pmatrix} + \begin{pmatrix} 0 & wi \\ wi & 0 \end{pmatrix} \\ &= \begin{pmatrix} x+yi & z+wi \\ -z+wi & x-yi \end{pmatrix}. \end{aligned}$$

Now, notice that the $(2,2)$ entry is the complex conjugate of the $(1,1)$ entry, and the $(2,1)$ entry is the negative conjugate of the $(1,2)$ entry. This relationship is clearly from the chosen basis vectors, so it holds for the image of any quaternion:

$$q \longmapsto \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \qquad \text{where} \qquad q = \alpha + \beta j.$$

By now, matrices of this form should be familiar. If we divide by the norm, we recover the $\mathrm{SU}(2)$ matrices!

Proposition 1.85

The Lie group $\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \middle| \ \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$ is isomorphic (as a ring) to the ring of unit quaternions.

Recall that $\mathrm{SU}(2)$ is the 2×2 complex matrices of determinant one. By identifying $\mathbb{C}^2$ with $\mathbb{R}^4$, an element of $\mathrm{SU}(2)$ determines a determinant one 4×4 matrix, i.e., a rotation in $\mathbb{R}^4$. Rotations in $\mathbb{R}^4$ are determined by a rotation of the unit sphere $S^3 \subset \mathbb{R}^4$, which is determined by the image of the $(1,0,0,0)$ vector, which is a point on S^3. This suggests the following claim.

Proposition 1.86

$\mathrm{SU}(2)$ is diffeomorphic to S^3.

Exercise 1.87

Prove this proposition.

All together, we can conclude that $\mathrm{SU}(2) \cong \{q \in \mathbb{H} \mid |q|^2 = 1\} \cong S^3$.

We can now ask: what is $\mathbb{HP}^1$? We now have the tools to understand why $\mathbb{HP}^1 \cong S^4$.

Recall that one construction of the projective spaces $\mathbb{R}P^1$ and $\mathbb{C}P^1$ are as the one point compactifications of $\mathbb{R}$ and $\mathbb{C}$, respectively; since $\mathbb{H} \cong \mathbb{R}^4$ and $\mathbb{R}^4 \cup \{\infty\} \cong S^4$, we can conclude that $\mathbb{HP}^1 \cong S^4$!

Alternatively, recall that $\mathbb{R}P^n$ is most commonly constructed as the quotient of S^n by the antipodal map. By viewing the antipodal map as the action of the "unit circle" $\{\pm 1\} \subset \mathbb{R}$, and further identifying $\{\pm 1\} \cong S^0$, $\mathbb{R}P^n$ can be seen as the base space of a fibration $p : S^n \to \mathbb{R}P^n$ with S^0 fibres. This construction is easily generalized to $\mathbb{C}P^n$ and then $\mathbb{HP}^n$: let V be a vector space over the ring $k = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$, and let $U(V^d)$ be the unit circle in V^d.

When $k = \mathbb{R}$, V^{n+1} is $n+1$ real dimensions, so $U(V^{n+1}_{\mathbb{R}}) \cong S^{n+1-1} = S^n$. The unit circle in $\mathbb{R}$ is $\{\pm 1\} \cong S^0$. This recovers the fibration

$$S^0 \to S^n \to \mathbb{R}P^n.$$

When $k = \mathbb{C}$, V^{n+1} is $2(n+1)$ real dimensions, so $U(V^{n+1}_{\mathbb{C}}) \cong S^{2(n+1)-1} = S^{2n+1}$. The unit circle in $\mathbb{C} \cong \mathbb{R}^2$ is $\{e^{i\theta}\} \cong S^1$, so the fibration becomes

$$S^1 \to S^{2n+1} \to \mathbb{C}P^n.$$

Note that when $n = 1$, this recovers the standard Hopf fibration $S^1 \to S^3 \to S^2$.

When $k \cong \mathbb{H}$, V^{n+1} is $4(n+1)$ real dimensions, so $U(V^{n+1}_{\mathbb{H}}) \cong S^{4(n+1)-1} = S^{4n+3}$. The unit circle in $\mathbb{H} \cong \mathbb{R}^4$ is S^3, thus the fibration is

$$S^3 \to S^{4n+3} \to \mathbb{HP}^n.$$

Instantons on S^4

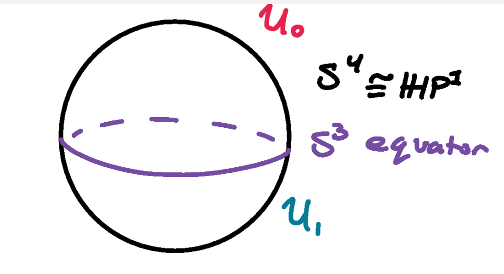
Mirroring our previous discussion of instantons on $\mathbb{C}P^1$, we'll view $S^4 = \mathbb{HP}^1$. $\mathbb{HP}^1$ admits two charts $U_0, U_1 \cong \mathbb{R}^4 \cong \mathbb{H}$, with the coordinate change function given by $q \mapsto \frac{1}{q}$.

What are SU(2) bundles over S^4?

SU(2) bundles correspond to maps $S^3 \to \mathrm{SU}(2)$, where S^3 is the region on which we glue U_0 to U_1. Since $\mathrm{SU}(2) \cong S^3$, SU(2) bundles over S^4 are classified by maps $S^3 \to S^3$, of which there are $\mathbb{Z}$-many (degree).

In terms of coordinates, some possible gluing maps are $g(q) = q^n/|q|^n$.

The $k = 1$ bundle corresponds to $g(q) = q$.



Part II: Instanton Floer Homology

Lecture 11 // Sept 29 // Instanton Floer Homology

The rest of the course will follow this general outline.

1. Broad overview of Instanton Floer Homology
2. Formal properties & applications
3. More detailed discussion of (1)
4. More recent papers

Instanton Floer Homology is a 3-manifold invariant. Originally, it was defined only for integer homology spheres, and we'll restrict to Y a $\mathbb{Z}\mathrm{HS}^3$ for simplicity, but the constructions we'll discuss can be extended.

The general procedure is as follows:

$$Y \rightsquigarrow C(Y) \rightsquigarrow H_*(C(Y)),$$

$\mathbb{Z}\mathrm{HS}^3$ Instanton chain complex Instanton Floer homology.

Where does $C(Y)$ come from? Two slogans:

1. ∞-dimensional Morse homology associated to the CS functional, i.e., there is a mysterious S^1-valued functional on an auxiliary space associated to Y (like $\mathcal{A}/\mathcal{G}$, but for Y^3):

$$\mathrm{CS} : \mathcal{A}/\mathcal{G} \longrightarrow \mathbb{R}/\mathbb{Z}.$$

   Whenever you have an $\mathbb{R}$-valued function, we take the Morse homology:

   - the critical points of CS become the generators of the chain complex,
   - the differentials are defined (roughly) by studying the gradient flow of CS.

2. The generators of $C(Y)$ are in correspondence with conjugacy classes of SU(2) representations of $\pi_1(Y)$. So $C(Y)$ actually comes from π_1.

Today, we'll broadly discuss (1) & (2), and why they're the same.

What is the Chern-Simons Functional

Take Y^3 a $\mathbb{Z}\mathrm{HS}^3$ (though nothing today depends on that). Fix an SU(2) bundle over Y. Since $c_2(Y) = 0$, there's only one such bundle, the trivial one $E = Y \times \mathbb{C}^2 \to Y$.

Just like before, we study $\mathcal{A} = \{\mathrm{SU}(2) \text{ connections on } E\}$ and $\mathcal{G} \curvearrowright \mathcal{A}$, $\mathcal{G} = \Gamma(\mathrm{Aut}(E))$. Since E is trivial, $\mathcal{G} \cong \mathrm{Maps}(Y, \mathrm{SU}(2))$.

Note that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$, since $\mathrm{SU}(2) \cong S^3$, and maps $Y \to S^3$ are determined by the degree $\deg(g) \in \mathbb{Z}$. $\mathcal{A}/\mathcal{G}$ = configuration space / gauge, the domain of CS.

Definition 2.1
The Chern-Simons Functional is a map

$$\mathrm{CS} : \mathcal{A}/\mathcal{G} \longrightarrow S^1, \qquad \mathrm{CS}(A) = \frac{1}{8\pi^2}\int_Y \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right).$$

By this, we mean first: fix a global trivialization of E. In this global trivialization, ∇ is represented by a matrix of 1-forms A, which is the input of the integral.

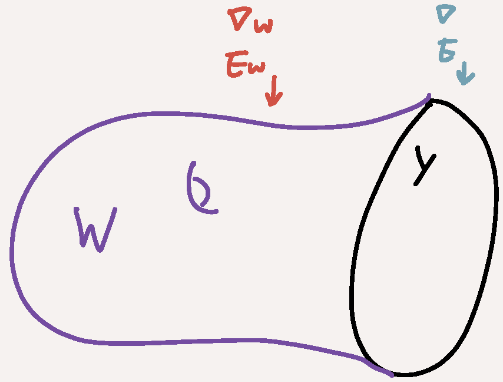
Exercise 2.2
If $g \in \mathcal{G}$ is a global gauge transformation, $\mathrm{CS}(g^*A) = \mathrm{CS}(A) + \deg(g)$.

This gives another way of saying that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$; each connected component corresponds to a trivialization, so this is saying that, up to homotopy, there are $\mathbb{Z}$-many trivializations of E. The value of the Chern-Simons Functional depends on which trivialization you choose. This definition is super explicit but not as insightful as the following alternate definition.

Definition 2.3
Let A be a connection on E. Choose any:

- smooth 4-manifold W^4 such that $\partial W \cong Y$,
- SU(2)-bundle E_W on W such that $E_W|_Y \cong E$,
- SU(2)-connection ∇_W such that $\nabla_W|_Y = \nabla$.

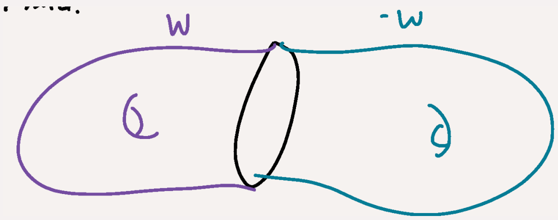
Then $\mathrm{CS}(\nabla) = \dfrac{1}{8\pi^2}\displaystyle\int_W \mathrm{tr}(F_{\nabla_W} \wedge F_{\nabla_W})$.



This definition follows from the definition of relative characteristic classes. But as-is, you should be skeptical that this is well defined. Let's choose a different W', $E_{W'}$, ∇'. Then

$$\mathrm{CS}(\nabla) = \frac{1}{8\pi^2}\int_W \mathrm{tr}(F_{\nabla_W} \wedge F_{\nabla_W}) \quad \text{and} \quad \mathrm{CS}(\nabla) = \frac{1}{8\pi^2}\int_{W'} \mathrm{tr}(F_{\nabla_{W'}} \wedge F_{\nabla_{W'}}).$$

If this definition is well-defined, then $\mathrm{CS}(\nabla) - \mathrm{CS}(\nabla') = 0$. To compute the difference, glue W and $-W'$ along Y to get an $\mathrm{SU}(2)$ bundle over a closed 4-manifold.



Then their difference becomes an element of $\mathbb{Z}$:

$$\frac{1}{8\pi^2}\int_W \mathrm{tr}(F_{\nabla_W} \wedge F_{\nabla_W}) - \frac{1}{8\pi^2}\int_W \mathrm{tr}(F_{\nabla_{W'}} \wedge F_{\nabla_{W'}}) = \langle c_2(E_\Sigma), [W \cup_Y -W'] \rangle \in \mathbb{Z}.$$

Thus different W's change $\mathrm{CS}(\nabla)$ by an integer, so $\mathrm{CS}(\nabla)$ is well-defined in S^1.

Remark 2.4

The bundles $E_W, E_{W'}$ over W and W' glue together to a well-defined bundle E_Σ over $W \cup_Y -W'$. The same is true for ∇_W and $\nabla_{W'}$, obtaining ∇_Σ. The Chern-Simons integral of this connection is that of a closed 4-manifold.

$$\mathrm{CS}(\nabla_\Sigma) = \frac{1}{8\pi^2}\int_{W\cup_Y -W'} \mathrm{tr}(F_{\nabla_\Sigma} \wedge F_{\nabla_\Sigma}) \in \mathbb{Z},$$

and splits as $\displaystyle\int_W \mathrm{tr}(F_{\nabla_\Sigma}|_W \wedge F_{\nabla_\Sigma}|_W) - \int_{W'} \mathrm{tr}(F_{\nabla_\Sigma}|_{W'} \wedge F_{\nabla_\Sigma}|_{W'})$.

However, for the connections ∇_W and $\nabla_{W'}$ over 4-manifolds with non-empty boundary, $\mathrm{CS}(\nabla_W) \notin \mathbb{Z}$. In our proof of well-definedness, we relied on some gluing procedure without verifying what that procedure actually is: this was incredibly sloppy. We should also ask whether this definition plays well with $\mathcal{G}$.

In order to glue E_W and $E_{W'}$ together, we must fix identifications $E_W|_Y \cong E_{W'}|_Y$. Since there is only one $\mathrm{SU}(2)$ bundle over Y, the trivial one, we could argue that $E_W, E_{W'}$ are trivial bundles, thus E_Σ is trivial, so the difference is zero. While the first part of this is true, like we saw with $\mathbb{CP}^1$, two trivial bundles could be glued via some nontrivial boundary map, like $\mathbb{Z}^n$, which results in a nontrivial bundle. Even simpler, two trivial $\mathbb{R}$-bundles over the interval I could be glued via a twist, resulting in the Möbius band. The number of these different maps is the number of retrivializations of the bundle on the boundary. So we must ask: how many retrivializations of ∇_W are there? The answer is $\mathbb{Z}$, as we saw earlier, and these $\mathbb{Z}$-many gluings give $\mathbb{Z}$-many $\mathrm{SU}(2)$ bundles over W!

Claim 2.5. *Re-trivializing a bundle E_W over $\partial W \cong Y$ by composing with some $g \in \mathcal{G}$ changes $c_2(E_\Sigma)$ by $\deg g \in \mathbb{Z}$.*

Lemma 2.6

The two definitions agree.

Proof. This follows immediately from Stokes' theorem: $\left(\int_W d\beta = \int_{\partial W} \beta\right)$.

$$8\pi^2 \,\mathrm{CS}(A) = \int_Y \mathrm{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right) = \int_W d\left(\mathrm{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)\right).$$

Simplifying,

$$d\left(\mathrm{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)\right) = \mathrm{tr}\left(d(A \wedge dA) + \frac{2}{3}d(A \wedge A \wedge A)\right)$$

$$= \mathrm{tr}\left(dA \wedge dA - A \wedge d^2 A + \frac{2}{3}\left(dA \wedge A \wedge A - A \wedge dA \wedge A + A \wedge A \wedge dA\right)\right)$$

$$= \mathrm{tr}\left(dA \wedge dA + \frac{2}{3} dA \wedge A \wedge A - \frac{2}{3} A \wedge dA \wedge A + \frac{2}{3} A \wedge A \wedge dA\right)$$

$$= \mathrm{tr}(dA \wedge dA + 2\, dA \wedge A \wedge A) = \mathrm{tr}(dA \wedge dA + dA \wedge A \wedge A + A \wedge A \wedge dA)$$

$$= \mathrm{tr}\left((dA + A \wedge A) \wedge (dA + A \wedge A)\right) + \mathrm{tr}(A \wedge A \wedge A \wedge A) = \mathrm{tr}(F_A \wedge F_A),$$

since $\mathrm{tr}(A \wedge A \wedge A \wedge A) = 0$. Thus

$$\int_Y \mathrm{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right) = \int_W \mathrm{tr}(F_A \wedge F_A) = 8\pi^2 \left\langle c_2(E_\Sigma), [W \cup_Y -W'] \right\rangle. \qquad \square$$

Remark 2.7

Timeline: this all happened pre-Donaldson, post relative characteristic classes.

Just like in Donaldson's theorem, we want to study the critical points of CS, which we'll denote by $\mathrm{Crit}(\mathrm{CS}) \subset \mathcal{A}/\mathcal{G}$. What about interpretation #2?

Theorem 2.8

$\nabla \in \mathcal{A}/\mathcal{G}$ is a critical point of the Chern-Simons functional iff ∇ is flat, i.e., $F_A = 0$.

Proof.

$$\nabla \in \mathrm{Crit}(\mathrm{CS}) \iff \left.\frac{d}{dt}\right|_{t=0} \mathrm{CS}(\nabla + tb) = 0 \iff \left.\frac{d}{dt}\right|_{t=0} 8\pi^2\,\mathrm{CS}(\nabla + tb) = 0$$

$$\iff \int_Y \mathrm{tr}\left((A + tb) \wedge d(A + tB) + \frac{2}{3}\left((A + tb) \wedge (A + tb) \wedge (A + tb)\right)\right) = 0$$

Using $d(A + tB) = dA \wedge B - A \wedge dB$,

$$\iff \int_Y \mathrm{tr}\left(B \wedge dA + A \wedge dB + \frac{2}{3} B \wedge A \wedge A + \frac{2}{3} A \wedge B \wedge A + \frac{2}{3} A \wedge A \wedge B\right) = 0$$

$$\iff \int_Y \mathrm{tr}(2B \wedge dA + 2B \wedge A \wedge A) = 0 = 2\int_Y \mathrm{tr}(B \wedge F_A) = 0 \quad \forall B$$

$$\iff F_A = 0.$$

$$\operatorname{grad}_A \mathrm{CS} = *F_A.$$ □

Theorem 2.9

Critical points of the Chern-Simons functional modulo gauge are in bijection with SU(2) representations of $\pi_1(Y)$ modulo conjugation.

Definition 2.10

The character variety of a 3-manifold is

$$R(Y) = \{\mathrm{SU}(2) \text{ representations of } \pi_1(Y)\}/\sim,$$

where $\rho_0 \sim \rho_1$ iff $\rho_1 = g^{-1} \cdot \rho_0 \cdot g$.

In this language, $\mathrm{Crit}(\mathrm{CS}) \leftrightarrow R(Y)$. We now prove the theorem.

Proof. Taking the holonomy gives a map

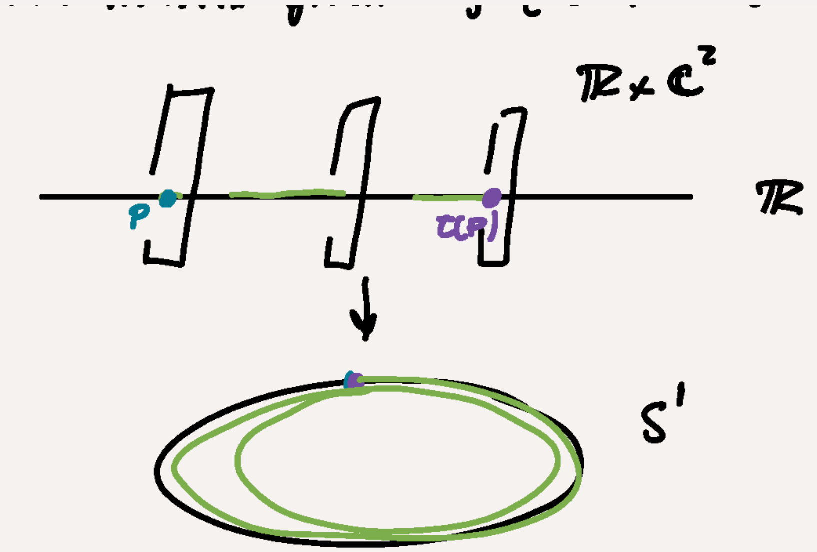
$$\{\text{flat connections}\} \to \{\mathrm{SU}(2) \text{ representations of } \pi_1(Y)\}.$$

Since $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$, this holonomy descends to a map on the quotient

$$\{\text{flat connections}\}/\mathcal{G} \to \{\mathrm{SU}(2) \text{ representations of } \pi_1(Y)\}/\sim \,= R.$$

It's not obvious this map is surjective. To show surjectivity, let $\rho : \pi_1(Y) \to \mathrm{SU}(2)$ be a representation of $\pi_1(Y)$. Construct an SU(2) bundle E and connection ∇ on E as follows:

1. Form $\widetilde{E} = \widetilde{Y} \times \mathbb{C}^2$, the trivial bundle over the universal cover of Y.
2. Quotient out $\widetilde{E}$ by identifying the fibres $p \times \mathbb{C}^2$ and $\tau(p) \times \mathbb{C}^2$ for $\tau \in \mathrm{Deck}(\widetilde{Y}) \cong \pi_1(Y)$ via the matrix given by $\rho(\gamma) \in \mathrm{SU}(2)$, where $\gamma \in \pi_1(Y) \leftrightarrow \tau$ between $p, \tau(p)$ (where γ lifts to τ).



This defines a bundle E over Y, as well as parallel transport on E by pushing down parallel transport on $\widetilde{E}$. By construction, this parallel transport will have holonomy ρ. This shows surjectivity.

$\square$

Lecture 12 // Oct 1 // The Chern-Simons Functional & Reducibles

Last time:
We defined the Chern-Simons functional

$$\mathrm{CS}(A) = \frac{1}{8\pi^2}\int_Y \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right)$$

and saw that $\mathrm{grad}_A \mathrm{CS} = *F_A$. We also interpreted $\mathrm{Crit}(\mathrm{CS})$ in two ways:

$$\mathrm{Crit}(\mathrm{CS}) = \{\text{flat connections modulo gauge}\} \leftrightarrow R.$$

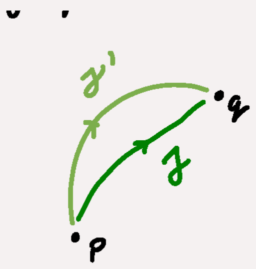
Today, we'll complete the proof of the theorem from last time.

Theorem (??)
Critical points of the Chern-Simons functional modulo gauge are in bijection with $\mathrm{SU}(2)$ representations of $\pi_1(Y)$ modulo conjugation.

Proof (continued). We must show injectivity. Let ∇_0, ∇_1 be two connections such that $\mathrm{hol}_p^{\nabla_0}$ and $\mathrm{hol}_p^{\nabla_1}$ are related by conjugation via some fixed $g_p \in \mathrm{SU}(2)$.

We want to show ∇_0 and ∇_1 are gauge equivalent. We construct a global gauge transformation from g_p as follows: for any q, choose a path γ from p to q. Then define

$$g_q = \mathrm{hol}^{\nabla_1}(\gamma) \circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}).$$



Is g_q well-defined? Yes: choose another γ' from p to q. Then $g_q' = \mathrm{hol}^{\nabla_1}(\gamma') \circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')$, and

$$\begin{aligned} g_q g_q'^{-1} &= \left(\mathrm{hol}^{\nabla_1}(\gamma) \circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right) \circ \left(\mathrm{hol}^{\nabla_1}(\gamma') \circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')\right)^{-1} \\ &= \left(\mathrm{hol}^{\nabla_1}(\gamma) \circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right) \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')^{-1} \circ g_p^{-1} \circ \mathrm{hol}^{\nabla_1}(\gamma')^{-1} \\ &= \mathrm{hol}^{\nabla_1}(\gamma) \circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}) \circ \mathrm{hol}^{\nabla_0}(\gamma') \circ g_p^{-1} \circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}'). \end{aligned}$$

Since $\mathrm{hol}^{\nabla_0}(\bar{\gamma}) \circ \mathrm{hol}^{\nabla_0}(\gamma') \in \mathrm{hol}_p^{\nabla_0}$, and $\mathrm{hol}_p^{\nabla_0}$ & $\mathrm{hol}_p^{\nabla_1}$ are related by $g_p \in \mathrm{SU}(2)$,

$$g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})) \circ \mathrm{hol}^{\nabla_0}(\gamma') \circ g_p^{-1} = \mathrm{hol}^{\nabla_1}(\bar{\gamma}) \circ \mathrm{hol}^{\nabla_1}(\gamma'),$$

so

$$g_q g_q'^{-1} = \mathrm{hol}^{\nabla_1}(\gamma) \circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}) \circ \mathrm{hol}^{\nabla_1}(\gamma') \circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}') = \mathrm{id}.$$

By construction, $\mathrm{hol}^{\nabla_0} = g_q^{-1} \circ \mathrm{hol}^0(\gamma) \cdot g_p$, so certainly the parallel transports of ∇_0, ∇_1 behave as if related by a gauge transformation. $\square$

The Instanton Chain Complex

Recall that the generators of the Instanton Chain Complex (ICC) are the critical points of the Chern-Simons functional, and the differential is defined by looking at the gradient flow of CS, $\mathrm{grad}_A(\mathrm{CS}) = *F_A$. But how can we do this on a three-manifold? The general slogan is that studying the gradient flow of CS on a three-manifold Y^3 is the same as studying SD/ASD Yang-Mills connections on $Y^3 \times \mathbb{R}$. There's a very precise connection between the three and four-manifold stories; this slogan is the first piece. Specifically, instead of studying (anti)-Self-Dual connections on closed X^4, we looked at the open 4-manifold $\mathbb{R} \times Y$. On $\mathbb{R} \times Y$, consider the trivial SU(2)-bundle $\mathbb{R} \times Y \times \mathbb{C}^2$. By the Künneth formula, $c_2(\mathbb{R} \times Y) = \sum_{i+j=2} c_i(\mathbb{R}) \otimes c_j(Y) = 0$, so this is only one.

Now let's fix a metric g_Y on Y, taking the product metric $dt^2 + g_Y$ on $\mathbb{R} \times Y$. We may use this to orient $\mathbb{R} \times Y$ by declaring a positively-oriented basis to be $\{dt, \underbrace{dx, dy, dz}_{Y \text{ directions}}\}$, with dt in the $\mathbb{R}$ direction.

Let's think about the bundle as a 1-parameter family of objects over Y. A gauge transformation g over $\mathbb{R} \times Y$ is a 1-parameter family of gauge transformations g_t: e.g., a gauge transformation on Y smoothly varying with t.

A connection (rather, a connection 1-form) on $\mathbb{R} \times Y$ is, in general, of the form $A = A_t + B_t\, dt$. This is achieved by separating the dt from dx, dy, dz in the 1-form, so A_t consists of one-forms in Y and B_t is a matrix of 0-forms. Note that A_t, B_t still vary in t, since the x, y, z coordinates are themselves functions of t: $x = x(t), y = y(t), z = z(t)$. Also note that B_t is not a connection on Y.

For each t, define the inclusion map $\iota_t : Y \to \{t\} \times Y \subset \mathbb{R} \times Y$; then A_t is $\iota_t^*(A)$. It will be convenient for us to define the following.

Definition 2.11
A is in temporal gauge if $B_t = 0$.

Lemma 2.12
Up to a gauge transformation on $\mathbb{R} \times Y$, every A can be put into temporal gauge.

Proof. We want to find a 1-parameter family of gauge transformations $\{g_t\}$ which kills the dt component. Applying the transformation law, g_t sends B_t to $g_t^{-1} B_t g_t^{-1} - g_t^{-1} dg_t$. Therefore g_t acts on A

via

$$A \xmapsto{\sim g_t} g_t^{-1} A g_t + g_t^{-1} dg_t.$$

Simplifying,

$$g_t^{-1} A g_t + g_t^{-1} dg_t = g_t^{-1}(A_t + B_t\, dt) g_t + g_t^{-1} dg_t = g_t^{-1} A_t g_t + g_t^{-1} B_t g_t\, dt + g_t^{-1} dg_t.$$

The dt component of $g_t^{-1} dg_t$ is $g_t^{-1}\frac{\partial}{\partial t} g_t\, dt$, so the dt component of $g_t^{-1} A g_t + g_t^{-1} dg_t$ is

$$g_t^{-1} B_t g_t\, dt + g_t^{-1} \frac{\partial}{\partial t} g_t\, dt.$$

Therefore we would like to show that for every t, there is a g_t such that

$$g_t^{-1} B_t g_t\, dt + g_t^{-1} \frac{\partial}{\partial t} g_t\, dt = 0.$$

In other words, we want to solve the 1-parameter family of ODEs:

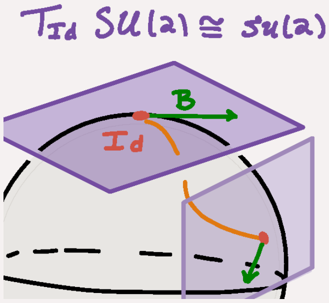
$$g_t^{-1} B_t g_t\, dt + g_t^{-1} \frac{\partial}{\partial t} g_t\, dt = 0 \iff B_t g_t\, dt + \frac{\partial}{\partial t} g_t\, dt = 0 \iff B_t g_t + \frac{\partial}{\partial t} g_t = 0 \iff \frac{\partial}{\partial t} g_t = -B_t g_t.$$

For each fixed $y \in Y$, we want that $\frac{\partial}{\partial t} g_t(y) = -B_t(y) g_t(y)$.

Let's interpret the right hand side. Note that $g_t(y)$ is a path in SU(2), and $B_t(y)$ is a time-varying element of the Lie algebra of SU(2), $\mathfrak{su}(2)$, which is the tangent space at the identity $T_{\mathrm{Id}}\,\mathrm{SU}(2)$. Explicitly, for a fixed t and y, $B_t(y)$ is a 2×2 matrix in $\mathfrak{su}(2)$, and $g_t(y)$ is a 2×2 matrix in SU(2).

We can get a new 2×2 matrix in SU(2) by right-multiplying $B_t(y)$ by $g_t(y)$:

$$B_t(y) \cdot g_t(y) = \begin{pmatrix} b_{11}(t,y) & b_{21}(t,y) \\ b_{12}(t,y) & b_{22}(t,y) \end{pmatrix} \cdot \begin{pmatrix} g_{11}(t,y) & g_{21}(t,y) \\ g_{12}(t,y) & g_{22}(t,y) \end{pmatrix}.$$



Doing so at each t, still fixing y, gives us the pushforward of $B_t(y)$ under right multiplication by $g_t(y)$, defining a vector field over all of SU(2).

Combining this with the left-hand side, if this vector field equals $\frac{d}{dt} g_t(y)$, the path must be an integral curve (e.g., a flowline) for this vector field. By ODE theory, this always exists; if the initial conditions are fixed, it is unique. □

Up to a gauge transformation, we can think of ∇ on $\mathbb{R} \times Y$ as $\{\nabla_t\}$ on Y.

We're now able to ask: what do the SDYM equations look like? First, the self-duality equation is

$F_A = *F_A$.

Let $A = A_t$ be in temporal gauge. Calculating $F_A = dA + A \wedge A$, note that it's not true that the 4-dimensional curvature is a 1-parameter family of 3-dimensional curvatures. Even though the connection itself has no dt component, the entries of this matrix are functions in all four variables, i.e., these entries are allowed to vary in t. This means that dA will have a $\frac{\partial}{\partial t}A_t\,dt$ component, which encodes how the curvature of Y changes in the time direction.

$$dA = dt \wedge \frac{\partial}{\partial t}A_t + \left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i}A_t\right) = -\frac{\partial}{\partial t}A_t \wedge dt + \underbrace{\left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i}A_t\right)}_{d^{(3)}A_t}$$

$$\Rightarrow F_A = -\frac{\partial}{\partial t}A_t \wedge dt + d^{(3)}A_t + A_t \wedge A_t.$$

For notational convenience, define $F_{A_t} := d^{(3)}A_t + A_t \wedge A_t$. Then $F_A = -\frac{\partial}{\partial t}A_t \wedge dt + F_{A_t}$.

To calculate $*F_A$, let's find the relationship between the three and four-dimensional Hodge stars. There are two main cases: if a 2-form has a time component, and if it doesn't.

1. $*^{(4)}(dx_1 \wedge dt) = -dx_2 \wedge dx_3 = -*^{(3)} dx_1$,
2. $*^{(4)}(dx_1 \wedge dx_2) = -dx_3 \wedge dt = -*^{(3)} (dx_1 \wedge dx_2) \wedge dt$.

This generalizes to the following relationship: let $i, j, k \in \{1, 2, 3\}$.

1. $*^{(4)}(dx_i \wedge dt) = -dx_j \wedge dx_k = -*^{(3)} dx_i$,
2. $*^{(4)}(dx_i \wedge dx_j) = -dx_k \wedge dt = -*^{(3)} (dx_i \wedge dx_j) \wedge dt$.

Thus $*^{(4)}F_A$ can be expressed using $*^{(3)}$ as

$$*^{(4)}F_A = *^{(4)}\left(-\frac{\partial}{\partial t}A_t \wedge dt + F_{A_t}\right) = *^{(4)}\left(-\frac{\partial}{\partial t}A_t \wedge dt\right) + *^{(4)}F_{A_t} = *^{(3)}\frac{\partial}{\partial t}A_t - *^{(3)}F_{A_t} \wedge dt.$$

In full, the SDYM equation becomes:

$$F_A = *F_A.$$

$$-\frac{\partial}{\partial t}A_t \wedge dt + F_{A_t} = *^{(3)}\frac{\partial}{\partial t}A_t - *^{(3)}F_{A_t} \wedge dt$$

$$\frac{\partial}{\partial t}A_t = *^{(3)}F_{A_t}.$$

For $A = A_t$ in temporal gauge, A is a SD YM connection iff A_t follows $\operatorname{grad}_B \mathrm{CS} = *F_B$. Similarly, A is ASD iff A_t is a flow of $-\operatorname{grad}_B \mathrm{CS}$.

Now that we've made sense of what it means to be a generator of the ICC, there's one technical point we must discuss. Just like in the proof of Donaldson's theorem, the space $\mathcal{A}/\mathcal{G}$ in the 3-dimensional setting is not a manifold, since there can be connections with nontrivial stabilizer. Recall that we proved that the stabilizer of a connection is the centralizer of the connection. This holds in the 3-d case too.

Subgroups of $\mathrm{SU}(2)$	Centralizer
I	$\mathrm{SU}(2)$
$-I$	$\mathrm{SU}(2)$
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
non-abelian	$\pm I$
$\mathrm{SU}(2)$	$\pm I$

Lemma 2.13

Flat connections with stabilizer larger than $\pm I$ correspond bijectively to abelian representations of $\pi_1(Y)$, i.e., those which factor through $H_1(Y;\mathbb{Z})$.

Definition 2.14

∇ is called reducible if $\mathrm{Stab}(\nabla)$ is larger than $\pm I$.

This is the first place we'll use that Y^3 is a $\mathbb{Z}\mathrm{HS}^3$. If Y is a homology sphere, there is only one reducible flat connection, which corresponds to the trivial representation into $\mathrm{SU}(2)$.

Remark 2.15

If Y is a $\mathbb{Q}\mathrm{HS}^3$, could have torsion. Computability of I breaks down based on how computable the character variety is, but also on how computable the differential is. The singular points aren't necessarily isolated, but they can be post perturbation.

Lecture 13 // Oct 6 // Defining Instanton Floer Homology

Last time:
We restricted to Y^3 a $\mathbb{Z}\mathrm{HS}^3$, and fixed the (trivial) SU(2) bundle E over Y. Let $\mathcal{A}^{\mathrm{irr}} \subset \mathcal{A}$ be the subset of connections with stabilizer $\pm I$.

Define

$$\mathcal{B} = \underbrace{\mathcal{A}}_{\text{SU(2) connections}} / \underbrace{\mathcal{G}}_{\text{gauge group}}, \qquad \mathcal{B}^* = \mathcal{A}^{\mathrm{irr}}/\mathcal{G}.$$

Then the Chern-Simons Functional is a map $\mathrm{CS} : \mathcal{B} \to \mathbb{R}/\mathbb{Z}$.

Definition 2.16 (IFH, irreducible complex)

$$C(Y) = \mathbb{F}\langle \text{crit points of CS on } \mathcal{B}^* \rangle.$$

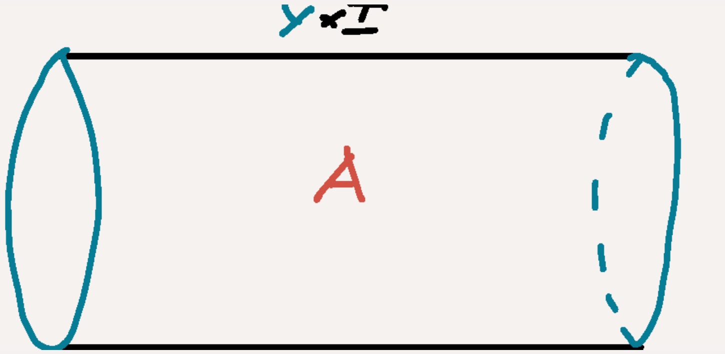
Remark 2.17
We have to perturb CS to ensure that the critical points are isolated. This perturbation happens in the same way one perturbs a function $f : M \to \mathbb{R}$ to be Morse.

Since $\mathcal{G}$ has $\mathbb{Z}$-many path components, $\pi_0(\mathcal{G}_0) = \mathbb{Z}$, so $\pi_1(\mathcal{B}^*) = \mathbb{Z}$. Now, take $x, y \in \mathrm{Crit(CS)}$ and let γ be a path from x to y in $\mathcal{B}$.

The set of homotopy classes of such paths is $\mathbb{Z}$.

Definition 2.18
The moduli space of γ-flow lines from x to y is

$$\mathcal{M}_\gamma(x,y) = \left\{ \begin{array}{l} \text{ASD connections on } Y \times I \\ \text{which limit to } x \text{ and } y \text{ on each} \\ \text{end and lie in homotopy class of } \gamma \end{array} \right\} \Big/ \text{gauge transformations on } Y \times I.$$



Note that there is a free $\mathbb{R}$ action on $\mathcal{M}_\gamma(x,y)$ by translation. If you've seen Morse theory, you know where this is going: we'll first quotient by $\mathbb{R}$, then count flowlines.

Definition 2.19

$\mathring{\mathcal{M}}_\gamma(x,y) = \mathcal{M}_\gamma(x,y)/\mathbb{R}$,

$$\partial x = \sum_y \sum_{\gamma \in S} \left| \mathring{\mathcal{M}}_\gamma(x,y) \right| y,$$

where S is the set of homotopy classes of paths from x to y for which $\dim(\mathring{\mathcal{M}}_\gamma(x,y)) = 0$.

It turns out that, in contrast to finite-dimensional Morse theory, the $\dim(\mathring{\mathcal{M}}_\gamma(x,y))$ depends on γ: different homotopy classes of path γ from x to y give different dimensional moduli spaces. In particular, the expected dimensions of the ASD moduli spaces differ by 8. Thus $C(Y)$ is a relatively $\mathbb{Z}/8$-graded complex. The relative grading difference between x and y is

$$\mathrm{gr}(x) - \mathrm{gr}(y) = \dim(\mathcal{M}(x,y)) \bmod 8.$$

To get an absolute $\mathbb{Z}/8$ grading on $\mathrm{CS}(Y)$, recall that for Y a $\mathbb{Z}\mathrm{HS}^3$, there is a unique (up to gauge) reducible flat connection on Y. Denote this connection by Θ, and define absolute gradings by counting flows into Θ.

Definition 2.20

The absolute grading of $x \in \mathrm{Crit}(\mathrm{CS})$ is

$$\mathrm{gr}(x) := \dim(\mathcal{M}(x,\Theta)) \bmod 8.$$

However, something goes wrong when we try to include Θ as a generator of our chain complex. To see this, let's add Θ. Since the boundary map ∂ is only defined for irreducible critical points, we must define new boundary maps for flows into and out of Θ. These are Donaldson's D_1 and D_2 maps.

Definition 2.21

$D_1 : C(Y) \to \mathbb{F}$ (think of as $\mathbb{F}\langle\Theta\rangle$),

$$D_1(x) = \sum_{\substack{\gamma \text{ from } x \text{ to } \Theta \\ \text{s.t. } \dim(\mathcal{M}(x,\Theta))=1}} \left| \mathcal{M}(x,\Theta)/\mathbb{R} \right|.$$

Definition 2.22

$D_2 : \mathbb{F} \to C(Y)$,

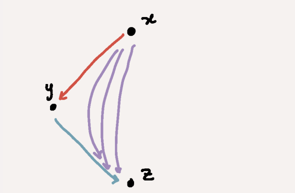
$$D_2(1) = \sum_y \left(\sum_{\substack{\gamma \text{ from } \Theta \text{ to } y \\ \text{s.t. } \dim(\mathcal{M}(\Theta,y))=1}} \left| \mathcal{M}(\Theta,y)/\mathbb{R} \right| y \right).$$

Notice that D_1 is supported on $C_1(Y) = \{x \in C(Y) \text{ s.t. } \mathrm{gr}(x) = 1\} \subset C(Y)$, since the only points which could flow into Θ must have grading 1. However, it turns out that D_2 has image in $C_{(4)}(Y)$. This is strange.

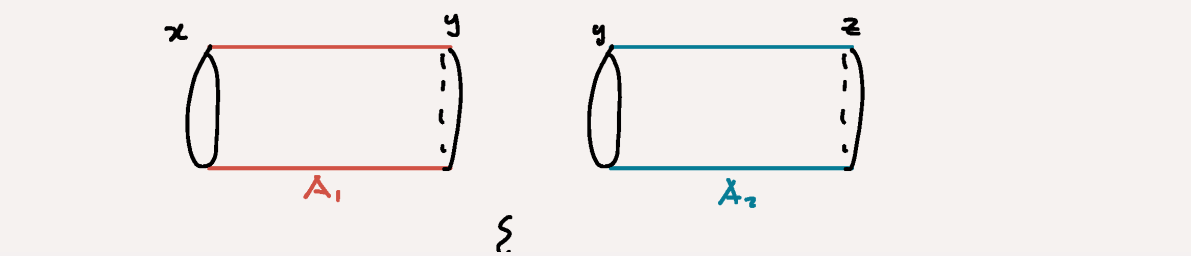
Let's go back to our definition of relative gradings. Why were these well-defined? The implicit claim here was that for $x, y, z \in \text{Crit}(\text{CS})$,

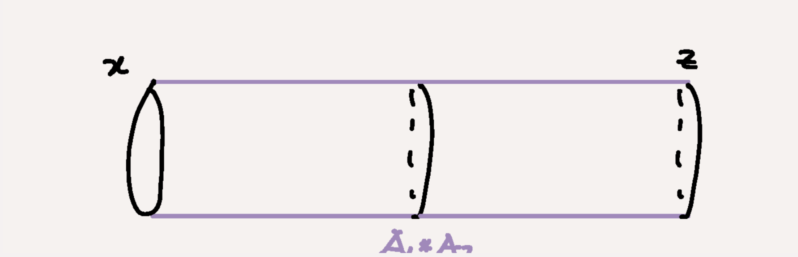
$$\dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) = \dim(\mathcal{M}(x,z)).$$

E.g., we can construct a flow from x to z by gluing flows from $x \to y$ to flows from $y \to z$.



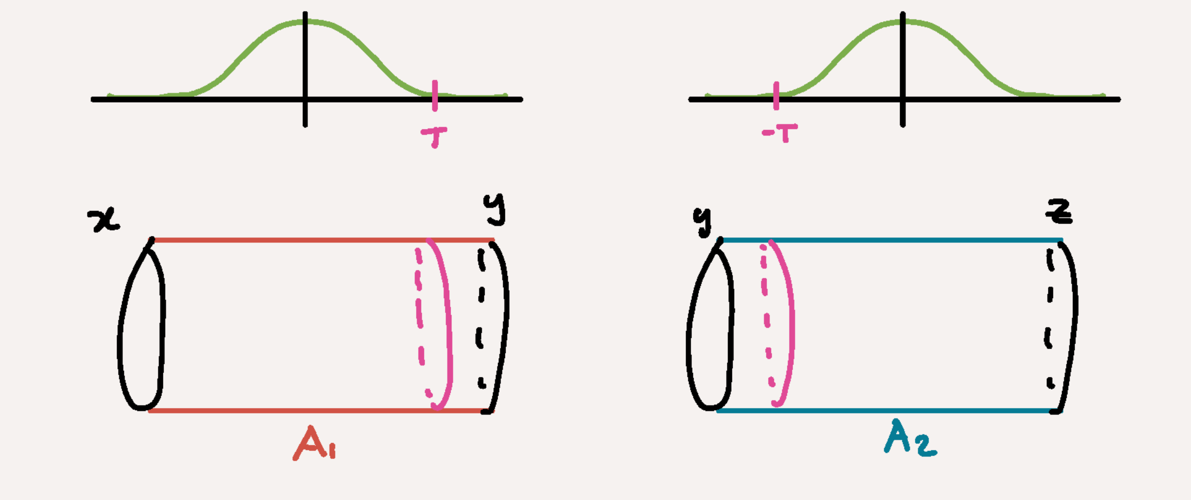
But something goes wrong in the Instanton Floer complex when $y = \Theta$. Say we believe in some gluing procedure. Then, given an ASD connection A_1 on $Y \times \mathbb{R}$ which limits to x as $t \to -\infty$ and y as $t \to \infty$, and similarly an A_2 from y to z, we'd want a connection $A_1 * A_2$ from x to z on $Y \times \mathbb{R}$ built from A_1 and A_2.

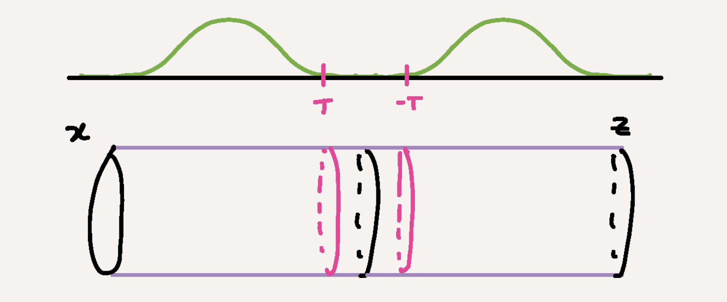




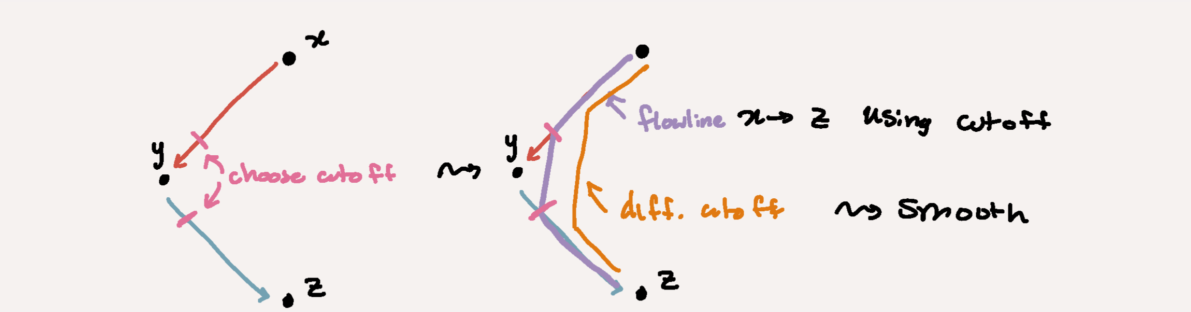
To "glue" A_1 and A_2 together at y to get something smooth, we'll need to add a decay function that ensures that after a certain time T,

$$f \circ A_1|_{[T,\infty)} = f \circ A_2|_{(-\infty,T]}.$$

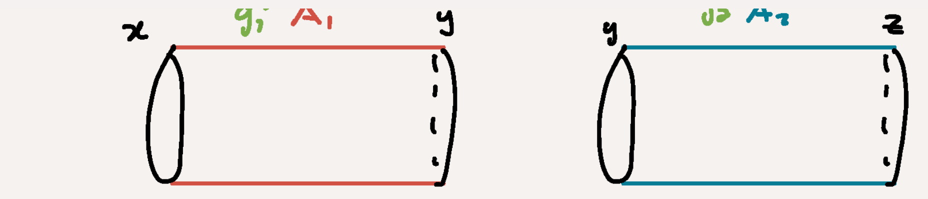

T
-T
x
y
y
z
A₁
A₂



This gives us a 1-dimensional gluing parameter T. In the flowline picture, the analog is building a flowline from x to z by choosing a cutoff parameter, following $x \to y$ until then,


x
y
z
choose cutoff
⇝
y
z
flowline x → z using cutoff
diff. cutoff
⇝ smooth

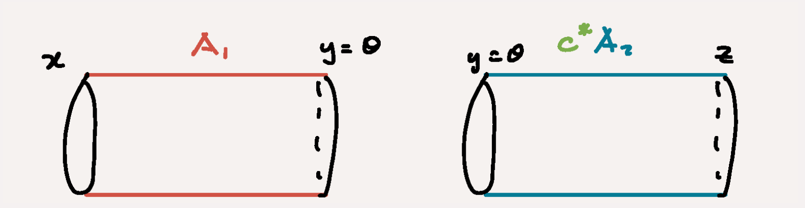
What goes wrong when y has large stabilizer? To glue, we can pick representatives on each end which are literally the same. But we can also compose with gauge transformations to get new gluings.


x
g₁* A₁
y
y
g₂* A₂
z

If $g_1^* A_1|_y = g_2^* A_2|_y$, then the connection on the composite cylinder is gauge-equivalent to the original connection glued via the identity.

But if $\mathrm{Stab}(y)$ is big, we get many more connections on $Y \times \mathbb{R}$ than we would expect!

$c =$ constant, non-identity gauge transformation on A_2.



Since y has big stabilizer, $c^*(A_2)|_y = A_2|_y$, so we can still glue to get a connection on the entire composite cylinder. However, $A_1 * c^* A_2$ is not gauge equivalent to $A_1 * A_2$, since any gauge transformation $c^* A_2 \to A_2$ must change A_1 as well. Therefore, we get a $\mathrm{Stab}(y)$'s worth of extra distinct connections on $A_1 * A_2$. Since $y = \Theta$, and $\mathrm{Stab}(\Theta) \cong \mathrm{SU}(2)$,

$$\dim(\mathcal{M}(x,z)) = \dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) + \dim(\mathrm{SU}(2)) = \dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) + 3.$$

This exactly matches what was claimed earlier: for $D_1 : x \to y$, $D_2 : y \to z$, we expect a grading difference of 2 for $x \to z$, but $1 - (-4) = 5 = 2 + 3$.

Another important fact is that in general, $D_2 \circ D_1 \neq 0$. This is in contrast with finite-dimensional Morse theory, where $\partial^2(x) = 0$ for all x. Let's recall why this is true.

$$\partial^2(x) = \partial(\partial(x)) = \partial\left(\begin{matrix}\sum_{y \in C_{k-1}} n(x,y)\, x \\ \#\text{flows } x \to y\end{matrix}\right) = \sum_{z \in C_{k-2}} \left(\sum_{y \in C_{k-1}} n(x,y)\, n(y,z)\right) y.$$

$\partial^2(x)$ breaks up as a sum of terms cz, where c is the number of broken flowlines from x to z.

$$\partial^2(x) = \sum \cdots + cz + \cdots,$$

The claim is that the coefficient c is always 0. Indeed, the moduli space $\mathcal{M}(x,z)/\mathbb{R}$ is a 2-dimensional manifold/$\mathbb{R}$, so it is an open 1-manifold. It admits a compactification, where the endpoints are broken flowlines. Thus the number of broken flowlines is even. Thus zero mod 2, and with slightly more work, one can show the signed count is also zero.

However, this relies on the fact that $\mathcal{M}(x,z)$ is a 2-dimensional manifold, which comes from the index difference of 2. Therefore this argument fails for Θ in IFH, so we can't add in the reducible and get a chain complex. But $D_1 \circ \partial = 0$ and $\partial \circ D_2 = 0$, so there's some hope of working around this. One such way is by augmenting.

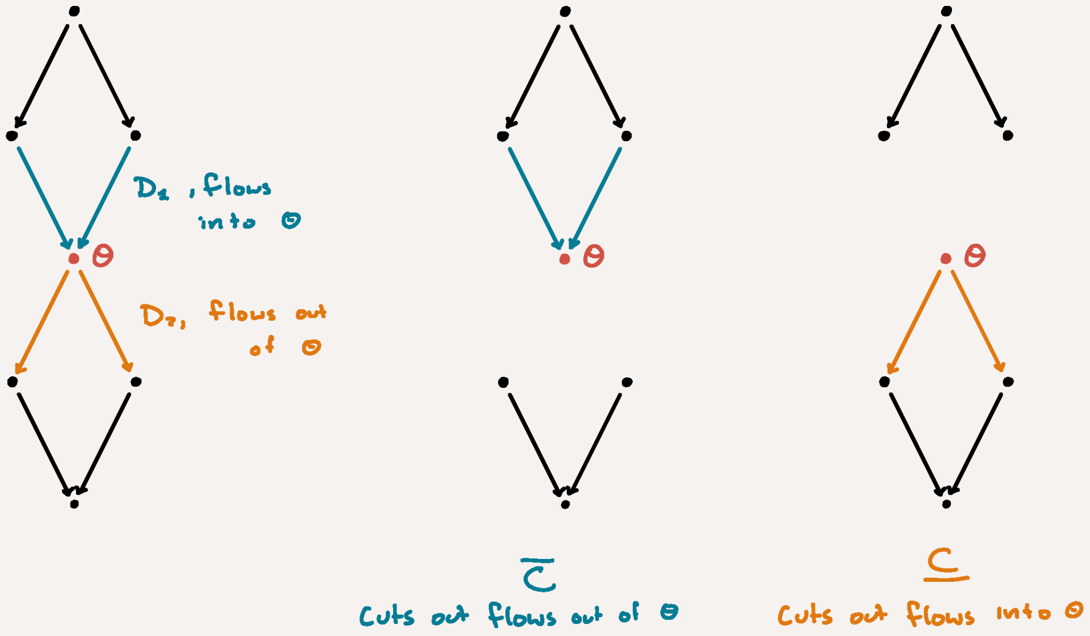
Note that $D_2(\Theta)$ is a cycle in $C(Y)$, and thus defines a homology class $[D_2(\Theta)] \in H_{-4}(C)$. D_1 defines a cohomology element since it vanishes on $\mathrm{im}(\partial)$, so $[D_1] \in H^1(C)$.

Given the irreducible complex C with D_1 and D_2, we can define the reducible complex by either precomposing D_2 or postcomposing D_1 with ∂, i.e., we only allow either flows into or out of Θ.

Definition 2.23

The "upper bar complex" is $\overline{C} = (C \oplus \mathbb{F}_{(0)},\ \overline{\partial} = \partial + D_1)$.

The "lower bar complex" is $\underline{C} = (C \oplus \mathbb{F}_{(-3)},\ \underline{\partial} = \partial + D_2)$.


D_1, flows into θ
θ
D_2, flows out of θ
θ
$\overline{C}$
Cuts out flows out of θ
θ
$\underline{C}$
Cuts out flows into θ

Lecture 14 // Oct 8 // Lifting to $\mathbb{Z}$-grading and the Filtered Chain Complex

Last time:

We defined the irreducible Instanton Chain Complex (ICC) $C(Y)$, which is a $\mathbb{Z}/8$ graded complex based at the reducible Θ. The chain complex acts strangely at Θ: the grading drops by 5 for flows passing through it, so there's an issue with broken flowlines and ∂^2 isn't 0.

We can get a chain complex with the reducible solutions by encoding the behavior at Θ into two maps.

$$D_1 : C_1 \to \mathbb{F}, \text{ which encodes flows into } \Theta,$$

$$D_2 : \mathbb{F} \to C_{(-4)}, \text{ which encodes flows out of } \Theta.$$

From these we can define two chain complexes for which $\partial^2 = 0$ by either forgetting flows into/out of Θ:

$$\overline{C} = C \oplus \mathbb{F}_0, \qquad \overline{\partial} = \partial + D_1,$$

$$\underline{C} = C \oplus \mathbb{F}_{(-3)}, \qquad \underline{\partial} = \partial + D_2.$$

Question 2.24. *Why do we care about the reducible(s)?*

Answer. The cobordism maps usually come from the D_1, D_2 maps.

$\mathbb{Z}$-Lifted version of IFH

We can lift the CS functional to be $\mathbb{R}$-valued (instead of $\mathbb{Z}$) by choosing a lift of $\mathcal{B}^*$ to the universal cover, $\widetilde{\mathcal{B}}^*$,

$$\pi_1(\mathcal{B}^*) \cong \mathbb{Z}, \qquad \pi_1(\widetilde{\mathcal{B}}^*) = 1.$$

Recall that $\pi_1(\mathcal{B}^*) \cong \mathbb{Z}$ since we are quotienting the space of connections, which is contractible, by $\mathcal{G}$, which has $\mathbb{Z}$-many path components, one in each degree. To get $\widetilde{\mathcal{B}}^*$, we must instead quotient by a subgroup of $\mathcal{G}$ which is $\mathbb{Z}$-times smaller, i.e., choosing one path component as a "basepoint." The most obvious one to choose is the degree 0 one.

Definition 2.25

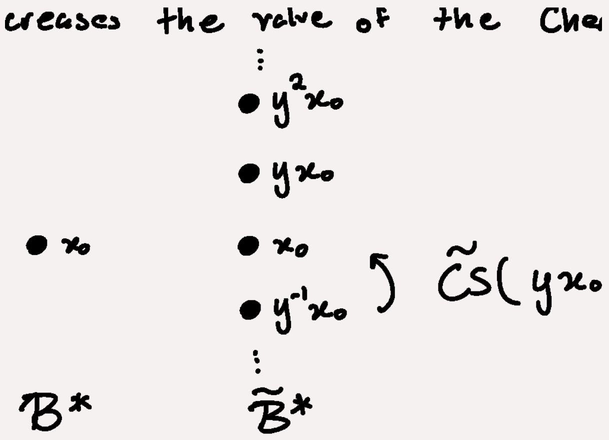
The based gauge group is the subgroup $\mathcal{G}_0 \leq \mathcal{G}$ of degree zero gauge transformations

$$\mathcal{G}_0 := \{g \in \mathcal{G} \mid \deg(g^{-1}) = 0\}.$$

By construction, $\pi_0(\mathcal{G}_0) = 1$. Like before, we define the configuration spaces $\widetilde{\mathcal{B}}$ and $\widetilde{\mathcal{B}}^*$ by

$$\widetilde{\mathcal{B}} = \mathcal{A}/\mathcal{G}_0, \qquad \widetilde{\mathcal{B}}^* = \mathcal{A}^{\text{irred}}/\mathcal{G}_0.$$

Note that $\widetilde{\mathcal{B}}^*$ is a $\mathbb{Z} : 1$ cover of $\mathcal{B}^*$, so the Chern-Simons functional $\mathrm{CS} : \mathcal{B}^* \to \mathbb{R}/\mathbb{Z}$ lifts to a $\mathbb{R}$-valued map $\widetilde{\mathrm{CS}} : \widetilde{\mathcal{B}}^* \to \mathbb{R}$. Repeating the IFH construction, we obtain a $\mathbb{Z}$-graded complex over $\mathbb{F}[y^{\pm 1}]$, where y is in degree 8 and multiplying by y encodes switching between lifts, so increases the value of the Chern-Simons functional by one.



$$\widetilde{\mathrm{CS}}(y \cdot x_0) = \widetilde{\mathrm{CS}}(x_0) + 1.$$

We will also refer to the $\mathbb{Z}$-lifted version as $C(Y)$.

$$\overline{C} = C \oplus \mathbb{F}[y^{\pm 1}]_{(0)}, \qquad \underline{C} = C \oplus \mathbb{F}[y^{\pm 1}]_{(-3)}.$$

By trivializing the bundle, we get a connection for which $\widetilde{\mathrm{CS}}$ vanishes; we take this to be the lift of Θ.

This filtration also descends to one on $\underline{C}$ and $\overline{C}$.

Definition 2.26

A filtered chain map $f : C_1 \to C_2$ is a chain map which is non-increasing with respect to the filtration, i.e., $f(C_1^s) \subseteq C_2^s \ \forall s$.

Definition 2.27

A filtered chain homotopy H between filtered chain maps f and g is a chain homotopy for which $H(C_1^s) \subset C_2^s \ \forall s$. If H exists for f, g, f and g are filtered chain homotopic.

We claim that the filtered homotopy equivalence class (FHEC) of C is an invariant of Y^3. To show this, we ask: what must we keep track of? What auxiliary data was used in our definition of $C(Y)$?

Well, the first thing we had to do for any bundle stuff was choosing a metric on Y. The critical points of $\mathrm{CS}(Y)$ don't depend on g: the point is that they're inherently topological, e.g., counting $\mathrm{SU}(2)$ representations of π_1. However, the differential does depend on g, since to even define a gradient-like vector field, we need a metric. Moreover, the definition uses the Hodge star! $\mathrm{grad}_A \mathrm{CS} = *_g F_A$.

So strictly speaking, the complex does depend on g a priori. Thus we must check that changing g doesn't change the FHEC.

There is a second thing. If CS doesn't have discrete critical points, we must perturb the CS functional to get discrete critical points. So far, we've brushed this fact under the rug; it's actually super annoying to deal with.

A random perturbation of CS functional (or even a general Morse function) will certainly change the value of CS at critical points. So we have no reason to think that any perturbation should play well with the CS filtration.

In reality, our definition of the IFC is incorrect. It's actually defined as a "limiting object." The idea is that we can find an arbitrarily small sequence of perturbations $\{\pi_i\}_{i=1}^{\infty}$ for which $\widetilde{\mathrm{CS}}$ has discrete critical points. For each perturbation, we get a filtered complex $C^{\pi_i}(Y)$. If we do this with enough regularity, as $\|\pi_i\| \to 0$, these complexes converge to some limiting object which we define to be $C(Y)$:

$$C^{\pi_i}(Y) \xrightarrow{\|\pi_i\|\to 0} C(Y).$$

However, there are many cases for which a perturbation is not needed, in which case the definition is true as given. For example, the Brieskorn homology spheres $\Sigma(p, q, r)$ with three singular fibres have discrete critical points without a perturbation. Conceptually, we'll discuss the complex as if no perturbation is needed.

The Chern-Simons Filtration

Definition 2.28

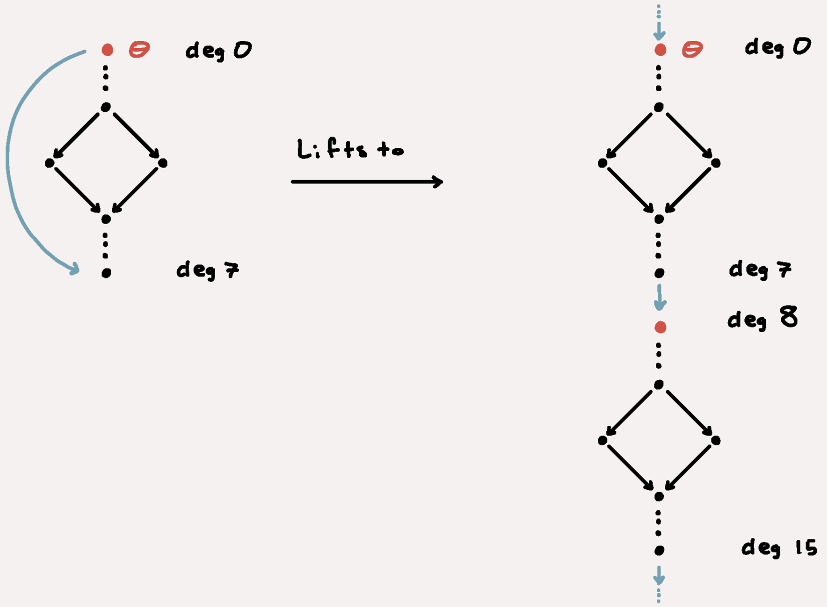
A $\mathbb{R}$-valued filtration on C is a collection of subcomplexes $\{C_s\}_{s\in\mathbb{R}}$ such that

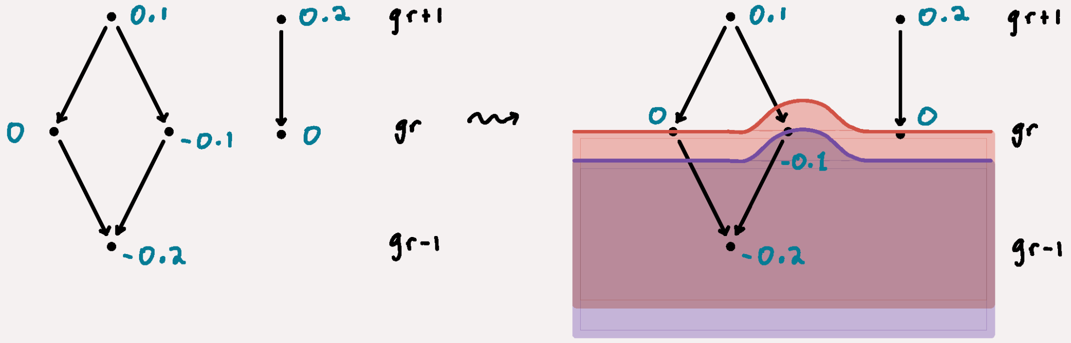
1. $C_r \subseteq C_s$ for $r \leq s$,
2. $\displaystyle\bigcup_s C_s = C$.

Given a filtered complex, we can get filtered subquotients $C^{[r,s]} = C^s/C^r$ for each pair $r \leq s$. Note that $[r, s]$ actually means $(r, s]$, so the generators $x \in C^{[r,s]}$ are the generators $x \in C$ with Chern-Simons value $r < \widetilde{\mathrm{CS}}(x) \leq s$.

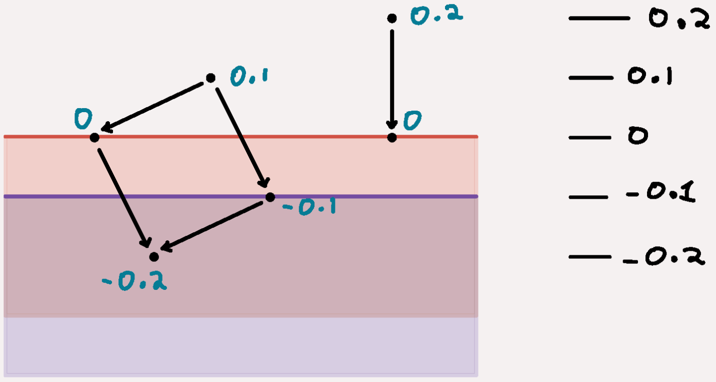
Observe that $\widetilde{\mathrm{CS}}$ gives C the structure of a filtered complex. For each generator (critical point), we can look at the value of the (perturbed) CS-functional at each x.

Diagrammatically, the filtration looks like the following. The values of CS for each generator are written in blue, and the complex is arranged vertically by $\mathbb{Z}$-grading.


θ
deg 0
deg 7
Lifts to
θ
deg 0
deg 7
deg 8
deg 15


0.1
0.2
gr+1
0
-0.1
0
gr
-0.2
gr-1
0.1
0.2
gr+1
0
0
gr
-0.1
-0.2
gr-1

If this complex was instead arranged vertically by Chern-Simons value, the filtration would look like horizontal lines:


0.2
0.1
0
0
-0.1
-0.2
0.2
0.1
0
-0.1
-0.2

Remark 2.29
The $\mathbb{Z}$-grading is independent from the CS value, but they both go down under the differential.

Remark 2.30

If $\mathcal{B}^*$ has nontrivial π_1, then the grading shift for flows which use this π_1 is larger than one, since the expected dimension for the flowline is higher. Lifting to $\widetilde{\mathcal{B}}^*$ "unwinds" this π_1, so gradings always drop by 1.

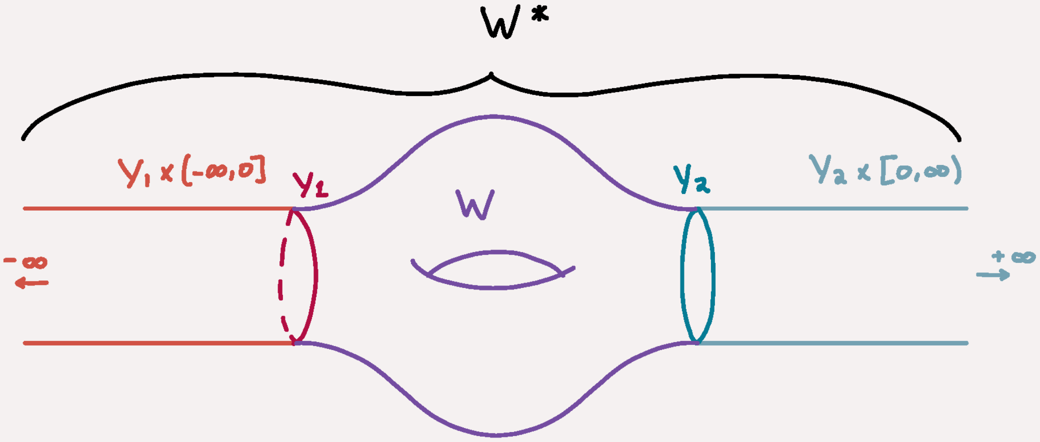
Cobordism Maps

If you've seen Floer theory before, you'll have heard of the so-called TQFT structure, e.g., "bordisms induce maps." This is formalized as follows.

Definition 2.31

A Floer homology theory F is said to have the structure of a TQFT if, for W a cobordism from Y_1 to Y_2, there is a cobordism map $F_W : C(Y_1) \to C(Y_2)$.

In our setting, let W be a negative-definite cobordism with $b_1(W) = 0$. From W, we construct the "cylindrical end manifold" W^* by attaching $Y_1 \times (-\infty, 0]$ and $-Y_2 \times [0, \infty)$ to their respective ends.



Consider the trivial bundle $W^* \times \mathbb{C}^2$. Given $x \in C(Y_1)$ and $y \in C(Y_2)$, look at ASD connections on W^* which limit to x and y (on the left and right, respectively), modulo gauge transformations on W^*. As with ∂, the space of connections on W^* modulo gauge on W^* has many different topological components. We can think about this as follows.

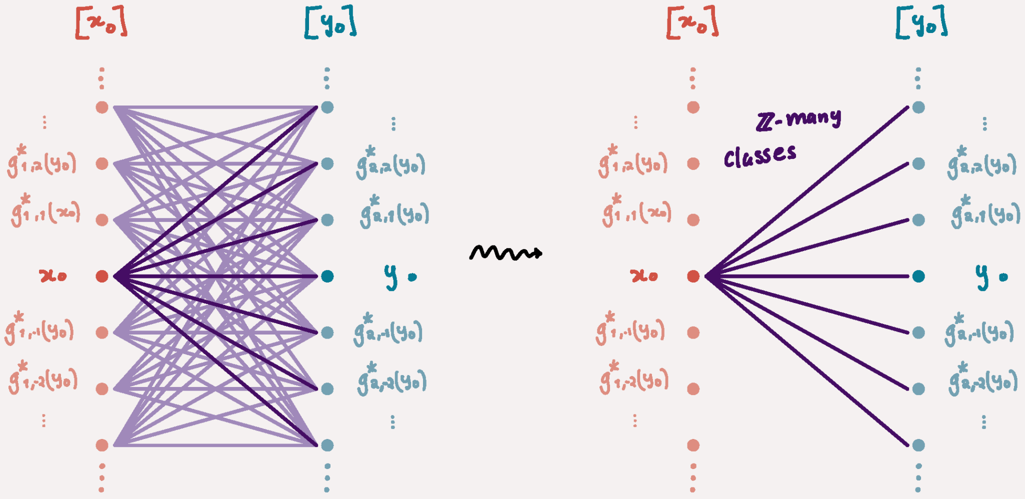
On Y_1 and Y_2, x and y are really just representatives of gauge equivalence classes $[x_0]$ and $[y_0]$ of connections on Y_1 and Y_2.

If $g^* y_0 \in [y_0]$, then we also want to count connections limiting to x_0 and $g^* y_0$. As with before, we claim there is no gauge transformation on W^* sending a connection from $x_0 \to g^* y_0$ to one from $x_0 \to y_0$. In particular, if g is a gauge transformation on W^*, then $\deg(g|_{Y_1}) = \deg(g|_{Y_2})$.

Exercise 2.32

Prove the claim.

This shows that there are $\mathbb{Z}$-many topological classes of connections interpolating between x_0 and y_0, determined by the difference of degree $\deg(g_1) - \deg(g_2)$ for gauge transformations g_1 on Y_1 and g_2 on Y_2. Diagrammatically, this can be seen by quotienting by a vertical $\mathbb{Z}$-shift:


[x₀]
[y₀]
g*₁,₂(y₀)
g*₁,₁(x₀)
x₀
g*₁,₋₁(y₀)
g*₁,₋₂(y₀)
g*₂,₂(y₀)
g*₂,₁(y₀)
y₀
g*₂,₋₁(y₀)
g*₂,₋₂(y₀)
[x₀]
[y₀]
ℤ-many classes
g*₁,₂(y₀)
g*₁,₁(x₀)
x₀
g*₁,₋₁(y₀)
g*₁,₋₂(y₀)
g*₂,₂(y₀)
g*₂,₁(y₀)
y₀
g*₂,₋₁(y₀)
g*₂,₋₂(y₀)

Part III: Applications

Lecture 17 // Oct 20 // Proof of the NST theorem

Last time:

We defined the r_s-invariant for $s \in [-\infty, 0]$

$$r_s(Y) = -\inf_{r<0} \{r \mid \text{there exists a } \Theta\text{-supported } [r, -s] \text{ cycle}\}$$

which asks the question "in which filtered subquotients do we have a cycle of the form $\Theta + \ldots$"

We showed that the r_s-invariant had the following properties:

1. r_s-invariant is a piecewise constant, non-increasing function.
2. If there exists a negative-definite cobordism W (with $H_1(W) = 0$) from Y_1 to Y_2, then $r_s(Y_1) \leq r_s(Y_2)$. In particular, r_s is a homology cobordism invariant: if Y bounds a negative-definite 4-manifold W (with $H_1(W) = 0$), then $r_s(Y) = \infty$.
3. In addition, if $\pi_1(W) = 1$, then the inequality is strict (allowing $\infty \leq \infty$).

Today, we'll prove one of the main theorems of the NST paper.

Theorem 3.1 (NST)

Let $k \subseteq S^3$. Denote $Y_n = S^3_{1/n}(k)$, $n > 0$.

- If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are homology cobordant to each other.
- If $r_s(Y_1) < \infty$ for any s, then $\{Y_n\}_{n=1}^{\infty}$ is linearly independent in $\Theta^3_{\mathbb{Z}}$.

Remark 3.2

A-priori, it's not obvious that $\Theta^3_{\mathbb{Z}}$ is at all interesting. For many years after its definition, it was not known whether $\Theta^3_{\mathbb{Z}}$ is 0, or ∞-generated. It is: in particular, it contains a $\mathbb{Z}^\infty$ subgroup ($\bigoplus_{i=1}^{\infty} \mathbb{Z}_i$). The original proof was due to Furuta and Fintushel-Stern in the 80's. In some sense, this theorem is a generalization of theirs. Furuta and Fintushel-Stern had the same family, and also used Yang-Mills theory. (Modern language (filtered) allows them to generalize it.)

Remark 3.3

It's actually very difficult to study $1/n$ surgeries using Heegaard Floer (HF). There's no known invariant coming from HF which distinguishes ∞-ly many $1/n$ surgeries on a fixed knot.

To study linear combinations of Y_n, we must first understand the behavior of r_s under $\#$.

Theorem 3.4

For any $s \in [-\infty, 0]$ and $s_1, s_2 \in [-\infty, 0]$ such that $s_1 + s_2 = s$,

$$r_s(Y_1 \# Y_2) \geq \min\{r_{s_1}(Y_1) + s_2,\ r_{s_2}(Y_2) + s_1\}.$$

Perhaps the form of this theorem – i.e., it is an inequality, and takes a minimum over a sum of r_s's and s's – is strange. We'll first give a pseudo-proof of the inequality, since the reason why it's of such a strange form will become apparent during the proof.

Proof (sketch). We'll pretend that $\underline{C}(Y_1 \# Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2)$. This ($\#$ corresponds to $\otimes$) is the behavior one usually expects for complexes under connect sum.

The reducible in $\underline{C}(Y_1 \# Y_2)$ is $\Theta_1 \otimes \Theta_2$, and $\widetilde{\mathrm{CS}}(\alpha \otimes \beta) = \widetilde{\mathrm{CS}}(\alpha) + \widetilde{\mathrm{CS}}(\beta)$.

Now, fix s_1 and s_2. Suppose that we have a Θ-supported cycle (a "OSC") α for $\underline{C}(Y_1)$ in $[r_1, -s_1]$, and a Θ-supported cycle β for $\underline{C}(Y_2)$ in $[r_2, -s_2]$.

What "Θ-supported" means is that in $\underline{C}(Y_1)$, $\alpha = \Theta + \dots$. To say that α is a cycle in this subquotient is to say that (a) α is in the subquotient, i.e., $\widetilde{\mathrm{CS}}(\alpha) \leq -s$, and (b) $\widetilde{\mathrm{CS}}(\partial\alpha) < r_1$. In particular, $\partial\alpha$ need not be 0, but if it isn't, it's outside the subquotient. The same is true for β: $\widetilde{\mathrm{CS}}(\beta) \leq -s_2$ and $\widetilde{\mathrm{CS}}(\partial\beta) < r_2$.

Claim: $\alpha \otimes \beta$ is a $\Theta_1 \otimes \Theta_2$-supported cycle (a.k.a., Θ-supported in the $\otimes$).

We know that $\widetilde{\mathrm{CS}}(\alpha \otimes \beta) \leq (-s_1) + (-s_2) = -s$. So $\alpha \otimes \beta$ is in the subquotient. By the Leibniz rule,

$$\partial(\alpha \otimes \beta) = (\partial\alpha) \otimes \beta + \alpha \otimes (\partial\beta).$$

Individually, $(\partial\alpha) \otimes \beta < r_1 + (-s_2)$ and $\alpha \otimes (\partial\beta) < (-s_1) + r_2$. Therefore $\alpha \otimes \beta$ is a cycle in the subquotient $[\max\{r_1 + (-s_2),\ (-s_1) + r_2\},\ -s]$. This is where the formula appears.

Now, note that this is true for all Θ-supported cycles in these subquotients. When does this occur? By definition, $-r_{s_1}(Y_1)$ and $-r_{s_2}(Y_2)$ are the borders of when there exists a Θ-supported cycle in each respective subquotient. So the claim is that you can take any r_1 up to $-r_{s_1}(Y_1)$ and any r_2 up to $-r_{s_2}(Y_2)$. Thus we get a Θ-supported cycle in the subquotient

$$[\max\{-r_{s_1}(Y_1) + (-s_2),\ -r_{s_2}(Y_2) + (-s_1)\},\ -s]$$

$$\Rightarrow\ r_s(Y_1 \# Y_2) \geq \max\{r_{s_1}(Y_1) + s_2,\ r_{s_2}(Y_2) + s_1\}.$$

□

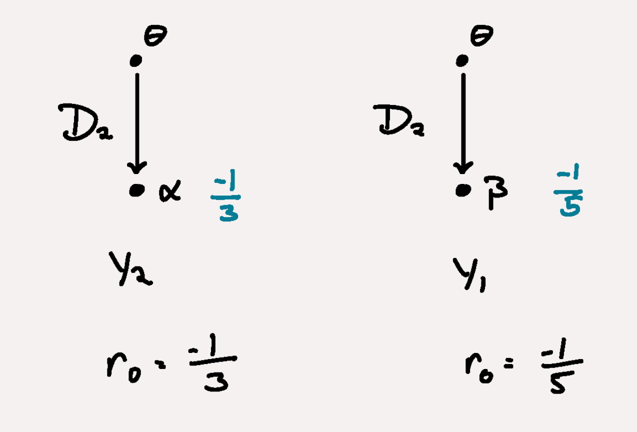
Corollary 3.5

$r_0(Y_1 \# Y_2) \geq \min\{r_0(Y_1), r_0(Y_2)\}$.

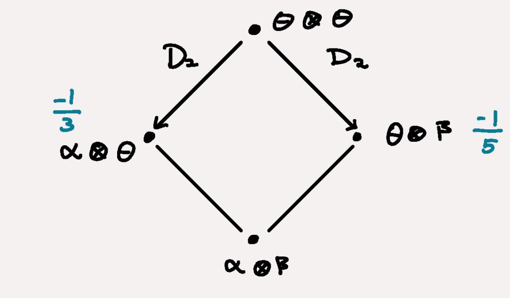
To see why this behavior – taking a minimum instead of adding – is more natural in this setting, let's do an example.

Example 3.6

Let the following be the $\underline{C}$ complexes for Y_1 and Y_2.



Here, r_0 is measuring the CS displacement of the D_2 arrow. The point is that if we pretend $\#$ is $\otimes$, the complex for $Y_1 \# Y_2$ looks like:



Notice that $D_2(\Theta)$ has 2 arrows, corresponding to the 2 summands in the Leibniz rule. Here, it's clear that to kill all the D_2's, you must take the minima of all of the summands.

Unfortunately, it's not actually true, in general, that $\underline{C}(Y_1 \# Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2)$. The rough reason is that it turns out that $\underline{C}(Y_1 \# Y_2)$ cannot be determined from $\underline{C}(Y_1)$ and $\underline{C}(Y_2)$ alone. To see why, let's look at the potential types of generators of $\underline{C}(Y_1 \# Y_2)$:

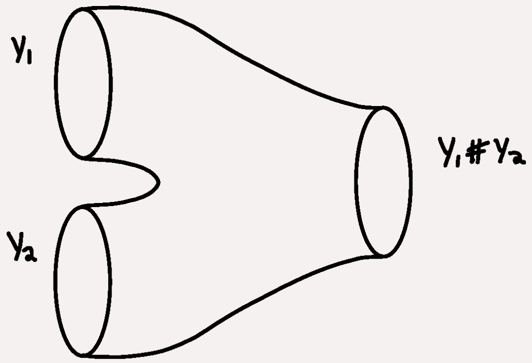
- $\Theta \otimes \Theta$ (red $\otimes$ red): this is red,
- $\alpha \otimes \beta$ (irred $\otimes$ irred),
- $\Theta \otimes \beta$ (red $\otimes$ irred),
- $\alpha \otimes \Theta$ (irred $\otimes$ red): these last three are irreducible in the tensor product.

The reason why the last two, which are combinations of reducible and irreducible generators, are irreducible in the product is as follows. Recall that generators correspond to SU(2) representations, and that $\pi_1(Y_1 \# Y_2) \cong \pi_1(Y_1) * \pi_1(Y_2)$. If we have SU(2) representations ρ_1 and ρ_2 of $\pi_1(Y_1)$ and $\pi_1(Y_2)$, and ρ_1 has trivial centralizer but ρ_2 doesn't, then $\rho_1 * \rho_2$ also has trivial centralizer. The intuition here is the same as for gauge transformations on the composite cylinder: even if some element commutes with ρ_2, it doesn't with ρ_1, so it doesn't commute with $\rho_1 * \rho_2$ either.

The issue arises because $\alpha \otimes \Theta$ and $\Theta \otimes \beta$ are irreducible: there is no reason to expect that there are no flows into/between them! In particular, we should be able to have maps from $\alpha \otimes \beta$ to $\alpha \otimes \Theta$. However, even if there is a D_1 map where $D_1(\beta) = \Theta$, in choosing the $\underline{C}$ complex, we threw these out! So you actually need both the D_1 and D_2 maps of each factor to calculate the $\underline{C}$ complex of the connect sum.

Remark 3.7
Being able to write down an explicit formula for $\underline{C}(Y_1 \# Y_2)$ as some tensor product is one motivation for why one may want to try harder to incorporate both D_1 and D_2 into the differential.

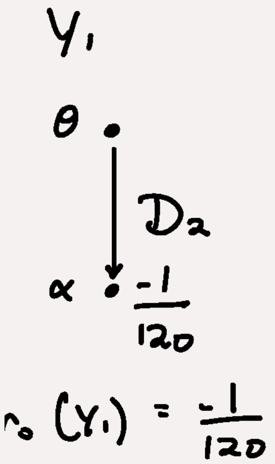
What salvages this is the fact that there is a cobordism map $\underline{C}(Y_1) \otimes \underline{C}(Y_2) \to \underline{C}(Y_1 \# Y_2)$. This cobordism is literally the pair of pants, from $Y_1 \sqcup Y_2$ to $Y_1 \# Y_2$.

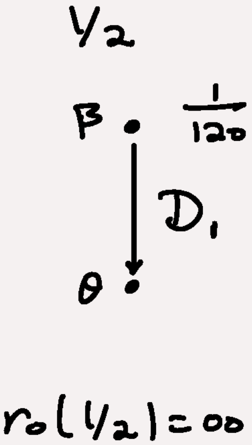


Thus far, we've assumed that the Y's are connected, which seems silly to note, but the distinction between connected and disconnected boundary really matters to the gauge group: they're qualitatively different. It turns out that we get bordism maps from disconnected → connected, but not the other way. But this means that the r_s invariant of the algebraic complex $\underline{C}(Y_1) \otimes \underline{C}(Y_2)$ bounds the r_s invariant of the complex we actually care about, $\underline{C}(Y_1 \# Y_2)$.

Question 3.8. *Is the inequality always strict?*

Answer. No! For example, take $Y_1 = -\Sigma(2,3,5)$ and $Y_2 = \Sigma(2,3,5)$. Then $Y_1 \# Y_2$ is homology cobordant to S^3. The $\underline{C}$ complex for each summand looks like





Since r_s is a homology cobordism invariant,

$$\infty = r_0(S^3) = r_0(Y\# -Y) \geq \min\{r_0(Y), r_0(-Y)\} \geq \min\left\{\frac{1}{120}, \infty\right\} \geq \frac{1}{120}.$$

We are now prepared to prove the theorem. We'll prove a more general algebraic statement.

Theorem 3.9

Let $\{Y_n\}_{n=1}^\infty$ be a sequence of $\mathbb{Z}\mathrm{HS}^3$'s such that

1. $\infty > r_0(Y_1) > r_0(Y_2) > \dots$
2. $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

Then the $\{Y_n\}_{n=1}^\infty$ are linearly independent in $\Theta^3_\mathbb{Z}$.

Proof. Assume not: suppose $\sum_{i\leq m} n_i Y_i = 0$ in $\Theta^3_\mathbb{Z}$. Then

$$n_m Y_m = \sum_{i<m} n_i Y_i \qquad \text{(assume } n_m > 0\text{)},$$

so $r_0(n_m Y_m) = r_0\left(\sum_{i<m} n_i Y_i\right)$. Note that these n_i can be positive or negative. Iterating this inequality, we get that

$$r_0(n_m Y_m) = \dots \geq \min\{r_0(\pm Y_1), r_0(\pm Y_2), \dots\},$$

where the + or − in each ± is chosen based on $\mathrm{sign}(n_i)$.

On the other hand, since $Y_m = n_m Y_m \# (-n_{m-1}) Y_m$,

$$r_0(Y_m) = r_0\left(n_m Y_m \# (-n_{m-1}) Y_m\right) \geq \min\{r_0(n_m Y_m), r_0(-Y_m)\} \geq \min\{r_0(n_m Y_m), \infty\} = r_0(n_m Y_m),$$

since $r_0(-Y_m) = \infty$ by hypothesis. Thus $r_0(Y_m) \geq r_0(n_m Y_m)$, so

$$r_0(Y_m) \geq \min\{r_0(\pm Y_1), r_0(\pm Y_2), \dots\},$$

which contradicts the initial hypotheses. □

The approximate intuition for this result is that small and nonzero CS displacement of D_2 cannot lie in the span of large CS displacements.

Does this apply to our situation? Yes! Here's a slick topological argument, which concludes the proof of NST.

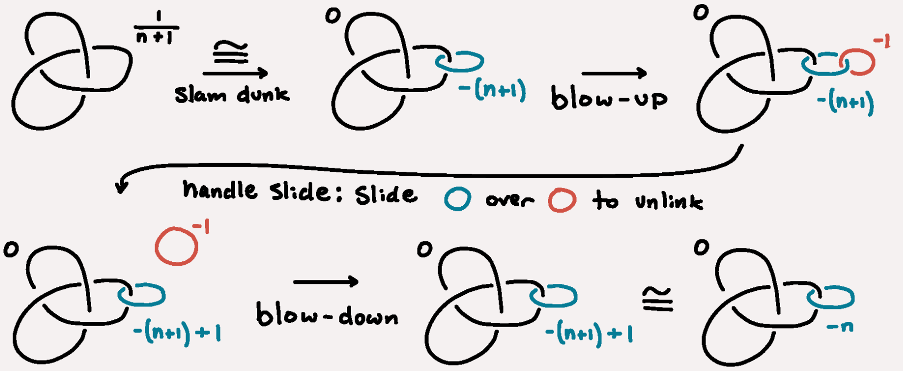
Theorem (Nozaki-Sato-Taniguchi (**??**))

Let $k \subseteq S^3$. Denote $Y_n = S^3_{1/n}(k)$, $n > 0$. If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are homology cobordant to each other.

Proof. We only need to show that the conditions of the previous theorem are satisfied. E.g., we need to verify that

1. $\infty > r_0(Y_1) > r_0(Y_2) > \dots$
2. $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

(1) holds from last time: we constructed a $\pi_1 = 1$ negative-definite cobordism from Y_{n+1} to Y_n: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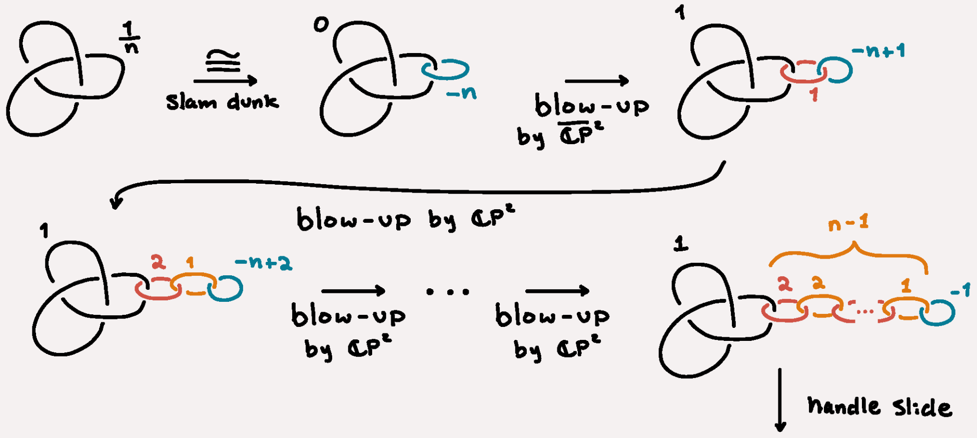
(2) follows from the fact that $S^3_{1/n}(k)$ bounds a positive-definite manifold if $n > 0$ and a negative-definite manifold if $n < 0$.

The positive-definite manifold can be constructed as follows ($n > 0$):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blow-up by $\mathbb{CP}^2$
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2
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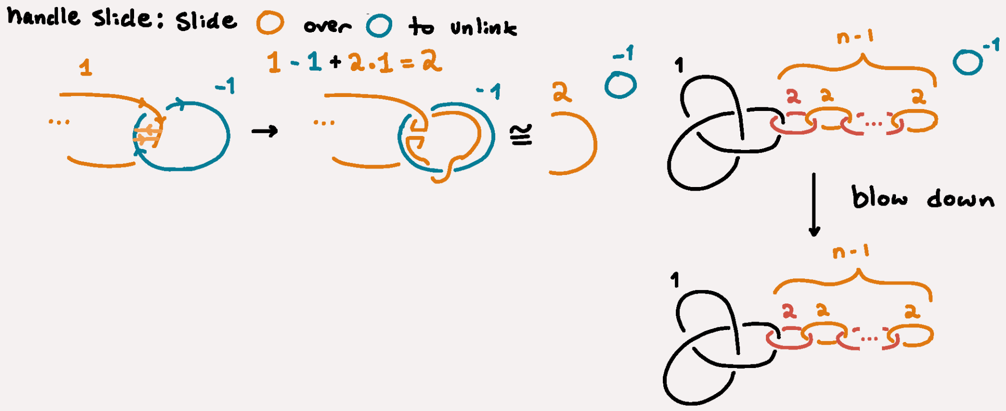
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2 2 ... 2

The intersection form is

$$\begin{array}{r} 1\,\{ \\ \\ n-1\,\{ \\ \\ \\ 1\,\{ \end{array}\begin{bmatrix} 1 & 1 & & & & 0 \\ 1 & 2 & 1 & & & \\ & 1 & 2 & 1 & & \\ & & 1 & \ddots & & \\ 0 & & & & 2 & 1 \\ & & & & 1 & 2 \end{bmatrix},$$

which we leave to the reader to verify has Smith normal form Id.

The same process can be applied when $n < 0$ by blowing up with $\overline{\mathbb{CP}}^2$'s. Since this process gives a negative-definite cobordism from $-Y_n$ to S^3, we can conclude that $r_0(-Y_n) = \infty$. ☐